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Variational Monte Carlo for Droplets of Neutral Atom Mixtures

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Abstract

Initially, a comprehensive overview of the Variational Monte Carlo (VMC) framework for quantum mechanical systems is provided. Subsequently, a binary bosonic mixture system, interacting via a Lennard-Jones potential and trapped inside a harmonic confinement, is studied and analyzed. The symmetrical case constitutes a good representation of a cluster of ${}^4\text{He}$ atoms. To this end, a VMC code is implemented and run. In spite of the suboptimal choice of trial wavefunction (flaws already identified in earlier work, such as McMillan's [12], are encountered here as well), the computed upper bounds to the sought-after ground-state energies are nonetheless sufficient to characterize the systems' qualitative behavior. In fact, the two-body benchmark shows the computed $N = 2$ energies exceed the true ground-state energies by between 0.3% and 9.6% depending on the system. Finally, the interaction-driven crossover from a confined trapped gas to a self-bound quantum liquid droplet, governed purely by the delicate balance of the Lennard-Jones potential, is uncovered. Specifically, clusters of $N = 6, 8, 12$ ${}^4\text{He}$ atoms manage to reach stability even in the limit of vanishing confinement ($\omega \rightarrow 0$).

Contents

Introduction	1
1 Variational Monte Carlo Fundamentals	3
1.1 Variational Principle	3
1.2 Variational Monte Carlo	4
1.2.1 Local Energy	4
1.2.2 The Vanishing-Variance Principle	4
1.3 Monte Carlo Integration	5
1.3.1 Markov Chain	5
1.3.2 Brute Force Metropolis Sampling	7
1.3.3 Importance Sampling and Fokker-Planck Dynamics	7
1.3.4 Umrigar Drift	8
1.4 Optimization of Parameters	8
1.4.1 The Loss Function	9
1.4.2 Newton-Raphson's Algorithm	9
1.4.3 Broyden-Fletcher-Goldfarb-Shanno (BFGS) Algorithm	10
1.4.4 Adaptive Moment Estimation (Adam) Algorithm	10
1.4.5 Statistical Energy Gradient and Variance Gradient	11
1.4.6 Nelder-Mead Algorithm	11
1.5 Error Analysis	12
1.5.1 The Automated Blocking Algorithm	12
2 The Physical System:	
Neutral Atom Mixtures	13
2.1 The Physical System	13
2.1.1 Historical Research Background	13
2.1.2 Formal Definition of the System	13
2.2 Trial Wavefunction	14
2.2.1 Local energy	14
2.2.2 Divergence cancellation	15
2.2.3 Dealing with different flavors	16
2.3 Alternative methods to evaluate local kinetic energies	17
2.4 Spatial Probability Densities	17
2.4.1 One-Body Density	18
2.4.2 Pair Correlation Function	19

3	Implementation	21
3.1	The Code's Objective and Structure	21
3.1.1	Summary of the Main Routines	21
3.1.2	The Idea behind the Implemented Classes	22
3.2	Implementation Choices	22
3.2.1	Dealing with Different Flavors and Automating the Solvers	22
3.2.2	Optimization of Parameters	23
3.2.3	Learning Rate Reduction on Plateau	24
3.2.4	Multi-threading & Blocking	24
4	Execution, Data and Results	27
4.1	Parameter Choice and Analytical Predictions	27
4.1.1	Lennard-Jones Potential	27
4.1.2	Harmonic Potential	27
4.1.3	Two-body Systems	28
4.2	Non-Interacting System	30
4.3	Interacting Systems	34
4.3.1	Strong confinement	34
4.3.2	Weak confinement	43
4.3.3	No confinement	48
	Conclusions	53
A	Appendix	55
A.1	Heuristical Analysis of the Confined Interacting Systems	55
	Bibliography	58
	List of Figures	60
	List of Tables	62

Introduction

Solving the Schrödinger equation for interacting many-body systems is generally impossible analytically and standard numerical integration methods suffer from an exponential scaling in computational cost as the number of particles increases. To overcome this limitation, stochastic approaches such as Quantum Monte Carlo methods are widely employed to estimate the properties of complex quantum systems.

In this thesis, we investigate the structural properties of a confined 3D bosonic neutral atom mixture interacting via a Lennard-Jones potential. In principle, such a system can model the extensively-studied ^4He atoms. Here we choose to extend the model to binary mixtures; that is, mixtures of two different species of atoms.

To this end, we develop a modular, Variational Monte Carlo engine capable of optimizing a Jastrow-correlated trial wavefunction. By tuning the cross-species Lennard-Jones interactions, we computationally explore the physics of the mixture, demonstrating the competition between kinetic energy, short-range hard-core repulsion, and long-range attraction, which, in the symmetric case, ultimately drives the system through a crossover from a trapped quantum gas to a self-bound quantum liquid droplet.

This thesis is the continuation and extension of the preceding Projects [1] and [2], and is structured as follows:

Chapter 1 : A comprehensive summary of the theory underpinning the Variational Monte Carlo method is presented. Section 1.4 specifically explores notable automated optimization algorithms tailored for energy minimization.

Chapter 2 : The physical system of interest is introduced and analyzed from an analytical perspective. Key mathematical quantities—crucial for the implementation of the chosen trial wavefunction—are explicitly derived, and relevant physical observables to be sampled via the Monte Carlo architecture are discussed.

Chapter 3 : An overview of the C++ code’s architecture, along with the core implementation choices and design patterns, is provided. The full source code can be found in the GitHub repository at <https://github.com/DavideCacopardi/VMC-bachelor-thesis>.

Chapter 4 : The program is initially benchmarked against exactly solvable non-interacting scenarios. Subsequently, the numerical results for the interacting mixture are collected and thoroughly analyzed, uncovering the emergence of the transition to the self-bound liquid regime.

Chapter 1

Variational Monte Carlo Fundamentals

1.1 Variational Principle

Consider a quantum mechanical system. Let the spectral decomposition of the Hamiltonian operator H be

$$H = \sum_n E_n |n\rangle\langle n|, \quad (1.1)$$

where $n \in \mathbb{N}$ and we sorted the energies in ascending order (i.e., $E_0 \leq E_1 \leq E_2 \leq \dots$) without loss of generality. We can consider a generic state $|\psi\rangle$ and write the expectation value

$$\langle\psi|H|\psi\rangle = \sum_n E_n \langle\psi|n\rangle \langle n|\psi\rangle. \quad (1.2)$$

A lower bound to the last expression is found by exploiting the chosen sorting of the energies:

$$\langle\psi|H|\psi\rangle \geq E_0 \sum_n \langle\psi|n\rangle \langle n|\psi\rangle = E_0 \langle\psi|\psi\rangle. \quad (1.3)$$

We end up with the following well-known *variational principle*:

$$E_0 \leq \frac{\langle\psi|H|\psi\rangle}{\langle\psi|\psi\rangle}. \quad (1.4)$$

If the ground state $(E_0, |n\rangle)$ is unknown, one can take advantage of this inequality evaluating the upper bound $\mathcal{E} = \langle\psi|H|\psi\rangle / \langle\psi|\psi\rangle$ by making use of an educated guess of the *trial state* $|\psi\rangle$. More specifically, one can vary the parameters which define $|\psi\rangle$ in order to find a tighter upper bound, closer to the true ground-state energy E_0 .

Working in the quantum D -dimensional coordinate-space of a system composed of N bodies, one can rewrite the expression for the estimator as

$$\mathcal{E} = \frac{\int d\mathbf{R} \psi^*(\mathbf{R}) H \psi(\mathbf{R})}{\int d\mathbf{R} |\psi(\mathbf{R})|^2}, \quad (1.5)$$

where we defined $\mathbf{R} = (\mathbf{r}_0, \dots, \mathbf{r}_{N-1})$ as the $(N \times D)$ -dimensional position vector of the particles building up our system. A generic Hamiltonian may be expressed in the following

form:

$$H = \sum_{i=0}^{N-1} \left[-\frac{\hbar^2}{2m_i} \nabla_i^2 + V_i(\mathbf{r}_i) \right] + \sum_{i=0}^{N-1} \sum_{j=0}^{i-1} U_{ij}(\mathbf{r}_i, \mathbf{r}_j) \quad (1.6)$$

1.2 Variational Monte Carlo

As previously suggested, the trial wavefunction employed can be defined by a set of M parameters $\boldsymbol{\alpha} = (\alpha_0, \dots, \alpha_{M-1})$:

$$\psi_T = \psi_T(\mathbf{R}; \boldsymbol{\alpha}).$$

We would like to explore different methods to optimize the estimator $\mathcal{E}$ with respect to the parameters $\boldsymbol{\alpha}$, but in order to do that we must first find a way to evaluate $\mathcal{E}$ efficiently.

1.2.1 Local Energy

We can rewrite Eq. 1.5 as follows:

$$\mathcal{E} = \frac{\int d\mathbf{R} \psi_T^* H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} = \frac{\int d\mathbf{R} \psi_T^* \frac{\psi_T}{\psi_T} H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} = \frac{\int d\mathbf{R} |\psi_T|^2 E_L}{\int d\mathbf{R} |\psi_T|^2} = \int d\mathbf{R} p(\mathbf{R}, \boldsymbol{\alpha}) E_L(\mathbf{R}, \boldsymbol{\alpha}), \quad (1.7)$$

where we defined the *local energy*:

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = \frac{1}{\psi_T(\mathbf{R}, \boldsymbol{\alpha})} H(\mathbf{R}) \psi_T(\mathbf{R}, \boldsymbol{\alpha}) \quad (1.8)$$

and the quantum PDF (*probability distribution function*):

$$p(\mathbf{R}, \boldsymbol{\alpha}) = \frac{|\psi_T(\mathbf{R}, \boldsymbol{\alpha})|^2}{\int d\mathbf{R}' |\psi_T(\mathbf{R}', \boldsymbol{\alpha})|^2}. \quad (1.9)$$

Evaluating the integral in equation 1.7 when $N \gg 1$ becomes computationally expensive if naive integration methods, such as trapezoidal integration, are employed. A better strategy is to use Monte Carlo integration methods instead.

1.2.2 The Vanishing-Variance Principle

A profound and extremely useful property of the local energy is known as the Zero-Variance Principle. Suppose our trial wavefunction $\psi_T(\mathbf{R})$ matches an exact eigenstate of the Hamiltonian, $\psi_0(\mathbf{R})$, with eigenvalue E_0 . In this scenario, the local energy evaluates to:

$$E_L(\mathbf{R}) = \frac{H\psi_0(\mathbf{R})}{\psi_0(\mathbf{R})} = \frac{E_0\psi_0(\mathbf{R})}{\psi_0(\mathbf{R})} = E_0$$

Remarkably, the local energy becomes completely independent of the spatial configuration $\mathbf{R}$. As a direct mathematical consequence, the statistical variance of the local energy, $\text{Var}(E_L)$, strictly vanishes as the trial wavefunction more finely approximates $\psi_0(\mathbf{R})$. Recent work by Webber and Lindsey [16] provides a rigorous proof for this quantitative extension and calls it the Vanishing-Variance Principle.

1.3 Monte Carlo Integration

For simplicity we rewrite Eq. 1.7, which contains the integral we need to evaluate.

$$\langle H \rangle = \int d\mathbf{R} p(\mathbf{R}, \alpha) E_L(\mathbf{R}, \alpha) \quad (1.10)$$

We can restrict ourselves to the 1-dimensional case and drop the dependency from the parameters α :

$$\mathbb{E}[E_L] = \int dx p(x) E_L(x) \quad (1.11)$$

If we draw a sufficiently large number N_{MC} of samples x_i from the distribution $p(x)$ we can also write:

$$\mathbb{E}[E_L] = \int dx p(x) E_L(x) \approx \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} E_L(x_i). \quad (1.12)$$

This step is justified by the *Strong Law of Large Numbers*, as long as we extract our points independently from one another.

Although this theoretical framework is elegant, it relies on a crucial assumption: our ability to extract independent samples directly from the complex, multidimensional probability distribution $p(\mathbf{R}) = |\psi_T(\mathbf{R})|^2 / \int d\mathbf{R}' |\psi_T(\mathbf{R}')|^2$.

In practice, generating independent random configurations for a quantum many-body system is computationally unfeasible. The probability density is typically sharply peaked in highly localized regions of the configuration space. If we were to naively sample the space using a uniform random distribution, the vast majority of our extracted points would land in regions where $p(\mathbf{R}) \approx 0$. These points would consume computational resources but contribute almost nothing to the final sum. Furthermore, as we increase the number of particles N , the volume of the configuration space grows exponentially.

To overcome this, we must leave aside the requirement of independent sampling. We use Markov Chain Monte Carlo methods to generate a sequence of *correlated* configurations that are mathematically guaranteed to asymptotically follow our target distribution $p(\mathbf{R})$.

1.3.1 Markov Chain

For simplicity, we consider a 1-dimensional system composed of one particle, whose state (e.g., its position) ranges over a discrete domain. Let $|\psi, t_0\rangle$ be its initial state, the probability of finding the system in a state i at a time $t_0 + \varepsilon$ is given by

$$|\langle i | \psi, t_0; t_0 + \varepsilon \rangle|^2 = w_i(t_0 + \varepsilon) = \sum_j W_{ij} w_j(t_0) \quad (1.13)$$

where the sum ranges over the whole domain of possible states j and $W_{ij} = W(j \rightarrow i)$ is an unknown transition probability from j to i . We are clearly assuming that W is time-independent. We can write the last equation in matrix form:

$$w(t_0 + \varepsilon) = W w(t_0) \quad (1.14)$$

and see that after n time steps

$$w(t_0 + n\varepsilon) = W^n w(t_0). \quad (1.15)$$

We can spectrally decompose the stochastic matrix W , obtaining eigenvalues λ_i , and expand the state vectors w in its diagonal basis $\{\nu_i\}$:

$$w(t_0 + n\varepsilon) = \sum_i \alpha_i \lambda_i^n \nu_i. \quad (1.16)$$

Since W is a stochastic matrix, it has the following properties:

$$\forall i, \quad \sum_j W_{ij} = 1 \quad \wedge \quad |\lambda_i| \leq 1 \quad \wedge \quad \max_i \{|\lambda_i|\} = 1 \quad (1.17)$$

This means we can sort the eigenpairs (λ_i, ν_i) so that

$$1 = |\lambda_0| \geq |\lambda_1| \geq |\lambda_2| \geq \dots \quad (1.18)$$

Now, we need to add the assumption that W is irreducible (i.e., starting from any state i , it is possible to reach any other state j). This allows the Perron-Frobenius Theorem to assure the multiplicity of $\lambda = 1$ to be exactly 1. Because of our sorting, $\lambda_0 = 1$. Furthermore, we require W to be aperiodic so that only λ_0 has magnitude 1.

$$1 = \lambda_0 > |\lambda_1| \geq |\lambda_2| \geq \dots \quad (1.19)$$

We can then rewrite equation 1.16 as

$$w(t_0 + n\varepsilon) = \alpha_0 \nu_0 + \alpha_1 \lambda_1^n \nu_1 + \alpha_2 \lambda_2^n \nu_2 + \dots \quad (1.20)$$

If we now take the limit for $n \rightarrow \infty$, we can see that the steady state $w = \alpha_0 \nu_0 = Ww$ is reached. After enough time steps, we are now allowed to rewrite Eq. 1.13 as

$$w_i = \sum_j W(j \rightarrow i) w_j. \quad (1.21)$$

$W(j \rightarrow i)$ is still unknown to us, but we can build a model for it. For instance we can write

$$w_i = \sum_j [w_j T(j \rightarrow i) A(j \rightarrow i) + w_i T(i \rightarrow j) (1 - A(i \rightarrow j))] \quad (1.22)$$

where T are the probabilities of proposing a certain transition and A are the probabilities of accepting said transition; the two terms in the sum account for both the probability of *transitioning to* state i and the probability of *staying in* state i (i.e., *not transitioning elsewhere*). Simplifying the equation leaves us with

$$\sum_j w_j T(j \rightarrow i) A(j \rightarrow i) = \sum_j w_i T(i \rightarrow j) A(i \rightarrow j) \quad (1.23)$$

which is satisfied by a stronger assumption referred to as *detailed balance*:

$$w_j T(j \rightarrow i) A(j \rightarrow i) = w_i T(i \rightarrow j) A(i \rightarrow j) \quad (1.24)$$

We can now return to the more general configuration space in 3 dimensions where the system of N particles lives.

In the end, we have come to know

$$\frac{T(j \rightarrow i) A(j \rightarrow i)}{T(i \rightarrow j) A(i \rightarrow j)} = \frac{w_i}{w_j} = \frac{p(\mathbf{R}_i)}{p(\mathbf{R}_j)} = \frac{|\psi_T(\mathbf{R}_i)|^2}{\int d\mathbf{R} |\psi_T(\mathbf{R})|^2} \frac{\int d\mathbf{R} |\psi_T(\mathbf{R})|^2}{|\psi_T(\mathbf{R}_j)|^2} = \frac{|\psi_T(\mathbf{R}_i)|^2}{|\psi_T(\mathbf{R}_j)|^2} \quad (1.25)$$

and now we can choose different approaches.

1.3.2 Brute Force Metropolis Sampling

The simplest method to construct this proper Markov chain is the standard random-walk Metropolis algorithm we often refer to as "brute force" Variational Monte Carlo. In this approach, we start from an arbitrary initial configuration $\mathbf{R}$ and iteratively propose new configurations $\mathbf{R}'$ by shifting a single particle's coordinates by a random amount. Specifically, if a particle is currently at position $\mathbf{r}_i$, we propose a new position $\mathbf{r}'_i$ using a uniform random displacement:

$$\mathbf{r}'_i = \mathbf{r}_i + \delta \cdot (\mathbf{u} - 0.5) \quad (1.26)$$

where $\mathbf{u}$ is a vector of random numbers uniformly distributed in $[0, 1]$, and δ is a set maximum step length. Since this proposal method is perfectly symmetric, $T(\mathbf{R} \rightarrow \mathbf{R}') = T(\mathbf{R}' \rightarrow \mathbf{R})$ which simplifies the detailed balance equation down to:

$$\frac{A(\mathbf{R} \rightarrow \mathbf{R}')}{A(\mathbf{R}' \rightarrow \mathbf{R})} = \frac{p(\mathbf{R}')}{p(\mathbf{R})} \quad (1.27)$$

This places a constraint on the ratio of the acceptance probabilities, but it does not uniquely define them. To maximize the efficiency of the configuration-space exploration, we want to accept proposed moves as frequently as possible when they take us to regions of greater interest. Thus, we choose the largest possible values for A that strictly remain valid probabilities ($A \leq 1$).

If the proposed state is more probable than the current state ($p(\mathbf{R}') \geq p(\mathbf{R})$), we automatically accept the move by setting $A(\mathbf{R} \rightarrow \mathbf{R}') = 1$. If the proposed state is less probable, we set the reverse acceptance $A(\mathbf{R}' \rightarrow \mathbf{R}) = 1$, which forces the forward acceptance to exactly equal the probability ratio. Summing up:

$$A(\mathbf{R} \rightarrow \mathbf{R}') = \min \left(1, \frac{p(\mathbf{R}')}{p(\mathbf{R})} \right) \quad (1.28)$$

Algorithmically, at each step, we extract a uniform random number $u' \in [0, 1]$. If $u' \leq A$, the move is accepted and $\mathbf{R}'$ is added to the chain. If $u' > A$, the move is rejected, and the old configuration $\mathbf{R}$ is counted again.

While computationally cheap per step, this symmetric random walk is physically "blind". The acceptance ratio will heavily depend on the chosen step length parameter δ . Setting δ too high increases the probability of the proposed move being rejected because moves will likely be proposed into regions where the wavefunction approaches zero. On the other hand, opting for a smaller step length leads to highly correlated samples and a long thermalization time. Thus the standard trade-off is to tune δ so as to achieve a $\sim 50\%$ acceptance ratio, which still means that half of the proposal steps are "wasted".

1.3.3 Importance Sampling and Fokker-Planck Dynamics

To drastically improve the sampling efficiency, we introduce *Importance Sampling*. Instead of a blind random walk, we model the particles as undergoing diffusion driven by a drift force, governed by the Langevin equation.

We introduce a *quantum force* $\mathbf{F}_Q$, which actively pushes the particles towards regions where the probability density $|\psi_T|^2$ is maximized. The quantum force is defined as the logarithmic gradient of the probability density:

$$\mathbf{F}_Q(\mathbf{R}) = 2 \frac{\nabla \psi_T(\mathbf{R})}{\psi_T(\mathbf{R})} = 2 \nabla \ln(\psi_T(\mathbf{R})) \quad (1.29)$$

Using this force, the proposed move for a particle is no longer symmetric. It is updated using a normally distributed random variable and a deterministic drift ($\mathbf{v}_{\text{drift}}(\mathbf{r}_i) = \mathbf{F}_Q(\mathbf{r}_i)$):

$$\mathbf{r}'_i = \mathbf{r}_i + D\mathbf{v}_{\text{drift}}(\mathbf{r}_i)\Delta t + \boldsymbol{\eta}\sqrt{2D\Delta t} \quad (1.30)$$

where $D = \hbar^2/(2m)$ is the diffusion constant, Δt is the simulated time step, and $\boldsymbol{\eta}$ is a random vector drawn from a standard normal distribution $\mathcal{N}(\mu = 0, \sigma = 1)$.

Because the proposal is no longer symmetric, we must use the generalized *Metropolis-Hastings* acceptance rule to preserve detailed balance. We must explicitly include the Green's function $G(\mathbf{R} \rightarrow \mathbf{R}')$, which defines the transition probability of the Langevin equation:

$$G(\mathbf{R} \rightarrow \mathbf{R}') = \frac{1}{(4\pi D\Delta t)^{3N/2}} \exp\left(-\frac{(\mathbf{R}' - \mathbf{R} - D\mathbf{v}_{\text{drift}}(\mathbf{R})\Delta t)^2}{4D\Delta t}\right) \quad (1.31)$$

The new acceptance ratio A becomes:

$$A(\mathbf{R} \rightarrow \mathbf{R}') = \min\left(1, \frac{G(\mathbf{R}' \rightarrow \mathbf{R})}{G(\mathbf{R} \rightarrow \mathbf{R}')} \frac{|\psi_T(\mathbf{R}')|^2}{|\psi_T(\mathbf{R})|^2}\right) \quad (1.32)$$

This directed sampling approach ensures that large steps can be taken in the configuration space while maintaining a high acceptance ratio, significantly reducing the autocorrelation time of the Markov chain.

1.3.4 Umrigar Drift

If the probability distribution contains singularities, the quantities measured by employing the standard Monte Carlo techniques discussed so far may be strongly biased by the walkers never sampling the neighborhoods of singularities or, conversely, starkly disturbed by them in the rare scenario of them being sampled.

Because of this, one can apply a small correction to the previously stated equations: as proposed in Umrigar's work [15], the drift term of Eq. 1.30 can be modified into

$$\mathbf{v}_{\text{drift}}(\mathbf{r}_i) = \frac{\sqrt{1 + 2aD|\mathbf{F}_Q(\mathbf{r}_i)|^2\Delta t} - 1}{aD|\mathbf{F}_Q(\mathbf{r}_i)|^2\Delta t} \mathbf{F}_Q(\mathbf{r}_i), \quad (1.33)$$

where $a \in [0, 1]$ is an interpolation parameter. As noted by Umrigar [15], the standard drift relies on the assumption that the quantum force $\mathbf{F}_Q = 2\nabla\psi_T/\psi_T$ is constant over the time step. Near a node, however, it is much more accurate to assume that $\nabla\psi_T$ is constant instead. The parameter a interpolates between these two regimes, effectively bounding the displacement and preventing instabilities when $|\mathbf{F}_Q| \rightarrow \infty$.

1.4 Optimization of Parameters

So far we have studied how we can evaluate the expectation value $\langle H \rangle$, given a fixed parameter vector $\boldsymbol{\alpha}$. This means we can list an array of reasonable $\boldsymbol{\alpha}$, let our program run for each one of them and estimate the optimal parameter vector (i.e., the one that minimizes the energy $E = \langle H \rangle$) visually, by means of a grid-search plot. Naturally, this serves only as a rough estimate and becomes increasingly more difficult and unreliable when our parameter vector has more and more components.

Moreover, since E is evaluated stochastically, optimizing it directly can sometimes lead to numerical instabilities: an artificially lower mean energy might be achieved by inadvertently pushing the parameters into regions where the wavefunction exhibits unphysical behavior and the variance drastically explodes.

1.4.1 The Loss Function

Recalling the Vanishing-Variance Principle (Subsection 1.2.2), we know that the true ground state concurrently minimizes the energy and strictly yields a zero variance for E_L . To mitigate the previously mentioned instabilities, it is highly advantageous to guide our search optimizing a mixed loss function that reflects both physical requirements:

$$\mathcal{L}(\boldsymbol{\alpha}) = \langle E_L \rangle + w \text{Var}(E_L),$$

where w is a tunable weight parameter, whose dimensions are the reciprocal of energy¹. This formulation acts as a physical regularizer. The $\langle E_L \rangle$ term naturally drives the parameters toward the minimum energy, while the $w \text{Var}(E_L)$ penalty heavily discourages trial wavefunctions that produce wildly fluctuating local energies. The overall result is a much smoother optimization landscape, which prevents overfitting the stochastic noise and efficiently guides the search toward a trial state that closely mimics the true target eigenstate. It is important to point out that the chosen functional form for the trial wavefunction is fundamentally restricted and might not be able to represent the true ground state exactly. Because of this analytical limitation, the set of parameters that minimizes the expected energy does not generally coincide with the one that minimizes the variance. The introduction of the weight w thus represents a deliberate trade-off between achieving the lowest possible variational energy and obtaining a smoother, computationally more stable local energy profile.

Once the loss function to optimize is established, one can abandon the manual grid search and make use of automated optimization algorithms. A brief overview of the selected methods follows.

1.4.2 Newton-Raphson's Algorithm

Newton-Raphson's algorithm begins by fixing an initial educated guess for the parameter vector. Let $\mathbf{g}(\boldsymbol{\alpha}) := \nabla_{\boldsymbol{\alpha}} \mathcal{L}(\boldsymbol{\alpha})$. At any n -th guess $\boldsymbol{\alpha}_n$ for the parameter vector, we can Taylor-expand $\mathcal{L}(\boldsymbol{\alpha})$ in the neighborhood of $\boldsymbol{\alpha}_n$ and evaluate it at $\boldsymbol{\alpha} = \hat{\boldsymbol{\alpha}}$, the optimal parameter vector we are seeking.

$$\begin{aligned} \mathcal{L}(\hat{\boldsymbol{\alpha}}) &= \mathcal{L}(\boldsymbol{\alpha}_n) + \mathbf{g}_n^\top (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n) + \frac{1}{2} (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n)^\top A_n (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n) + O((\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_n)^3) = \\ &= \mathcal{L}_n + \mathbf{g}_n^\top \mathbf{b}_n + \frac{1}{2} \mathbf{b}_n^\top A_n \mathbf{b}_n + O(|\mathbf{b}_n|^3) \end{aligned} \tag{1.34}$$

¹While a dimensionally homogeneous loss function such as $\mathcal{L}' = \langle E_L \rangle + w' \sqrt{\text{Var}(E_L)}$ (with a dimensionless w' factor) might appear physically more natural, it introduces severe numerical instabilities for gradient-based optimization. The derivative of the standard deviation yields a gradient proportional to $[\text{Var}(E_L)]^{-1/2} \nabla \text{Var}(E_L)$. Since the variance approaches zero near the exact ground state, this derivative diverges. In a Monte Carlo setting, this divergence heavily amplifies the residual stochastic noise, causing gradient-based algorithms to inherently crash or diverge exactly as they approach the true minimum. Using the variance directly avoids this mathematical singularity, ensuring well-behaved gradients.

where we defined

$$\mathcal{L}_n := \mathcal{L}(\alpha_n), \quad \mathbf{b}_n := \hat{\alpha} - \alpha_n, \quad \mathbf{g}_n := \mathbf{g}(\alpha_n), \quad A_n := \left\{ \partial_{\alpha_i \alpha_j}^2 \mathcal{L} \right\}_{i,j} \bigg|_{\alpha_n}.$$

Now we differentiate both sides and set them to zero. By doing so, we find $\mathbf{b}_n = -A_n^{-1} \mathbf{g}_n$, which is Newton-Raphson's iterative proposal:

$$\alpha_{n+1} = \alpha_n - A_n^{-1} \mathbf{g}_n. \quad (1.35)$$

At this point we could iterate this method until $\|\alpha_{n+1} - \alpha_n\|_2 \leq \varepsilon$, where ε is a chosen precision (typically $\sim 10^{-5}$).

The major drawback of this approach is that evaluating the exact A_n^{-1} matrix is extremely expensive. Thus, we are interested in finding a way to estimate it; that is, we rewrite the proposal to the more general expression

$$\alpha_{n+1} = \alpha_n - \eta_n C_n \mathbf{g}_n, \quad (1.36)$$

where η_n is a scalar *learning rate*, and try to find a matrix C_n that well approximates the inverse Hessian A_n^{-1} while remaining computationally cheap.

1.4.3 Broyden-Fletcher-Goldfarb-Shanno (BFGS) Algorithm

Starting off with a principle similar to the one used in the one-dimensional secant method, the BFGS algorithm iteratively updates C_n without needing costly matrix inversions. Defining the parameter step $\mathbf{s}_n := \alpha_{n+1} - \alpha_n$ and the gradient change $\mathbf{y}_n := \mathbf{g}_{n+1} - \mathbf{g}_n$, the update rule for the inverse Hessian is given by:

$$C_{n+1} = \left(\mathbb{1} - \frac{\mathbf{s}_n \mathbf{y}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n} \right) C_n \left(\mathbb{1} - \frac{\mathbf{y}_n \mathbf{s}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n} \right) + \frac{\mathbf{s}_n \mathbf{s}_n^\top}{\mathbf{y}_n^\top \mathbf{s}_n}. \quad (1.37)$$

Note that the underlying convergence proof requires the loss function $\mathcal{L}(\alpha)$ to be convex; this condition is usually satisfied in a neighborhood of the local minimum for cases of major interest.

1.4.4 Adaptive Moment Estimation (Adam) Algorithm

In this optimization algorithm [9], running averages with exponential forgetting of both the gradients and the second moments of the gradients are used. It is the method of choice for most neural network trainings, as it is simple enough, highly efficient even in higher dimensions and sufficiently noise-forgiving for most applications. It still requires a lot of experimenting to find the *learning rate* parameter η that works best in each specific scenario. Starting off by fixing an initial educated guess for the parameter vector, the iteratively proposed parameter update is

$$\alpha^{(t)} := \alpha^{(t-1)} - \eta \frac{\hat{m}_\alpha^{(t)}}{\sqrt{\hat{v}_\alpha^{(t)} + \varepsilon}}, \quad (1.38)$$

where $t = 1, 2, 3, \dots$ is the iteration index, $\varepsilon \ll 1$ prevents numerical errors, and

$$\hat{m}_\alpha^{(t)} = \frac{1}{1 - \beta_1^t} \left[\beta_1 m_\alpha^{(t-1)} + (1 - \beta_1) \nabla_\alpha \mathcal{L}^{(t-1)} \right],$$

$$\hat{v}_{\alpha}^{(t)} = \frac{1}{1 - \beta_2^t} \left[\beta_2 v_{\alpha}^{(t-1)} + (1 - \beta_2) \left(\nabla_{\alpha} \mathcal{L}^{(t-1)} \right)^2 \right]$$

are the first and second moment.

1.4.5 Statistical Energy Gradient and Variance Gradient

In order to feed any gradient-descent based optimizer, such as BFGS or Adam, we must be able to efficiently compute the objective gradient $\mathbf{g}(\alpha) = \nabla_{\alpha} \mathcal{L}$.

Initially, we can evaluate the derivative of the expectation value of the energy analytically, exploiting the hermiticity of H , and obtain:

$$\frac{\partial}{\partial \alpha_i} \langle E_L \rangle = \frac{\partial}{\partial \alpha_i} \left(\frac{\int d\mathbf{R} \psi_T^* H \psi_T}{\int d\mathbf{R} \psi_T^* \psi_T} \right) = 2 \left(\left\langle E_L \frac{\partial \ln |\psi_T|}{\partial \alpha_i} \right\rangle - \langle E_L \rangle \left\langle \frac{\partial \ln |\psi_T|}{\partial \alpha_i} \right\rangle \right) \quad (1.39)$$

One can also rewrite this expression introducing the operator $O_{\alpha_i}(x) = \frac{\partial \ln |\psi_T|}{\partial \alpha_i}$:

$$\frac{\partial}{\partial \alpha_i} \langle E_L \rangle = 2(\langle O_{\alpha_i} E_L \rangle - \langle O_{\alpha_i} \rangle \langle E_L \rangle) = 2 \text{Cov}(O_{\alpha_i}, E_L) \quad (1.40)$$

This means that we can evaluate the gradient $\mathbf{g} = \nabla_{\alpha} \langle E_L \rangle + w \nabla_{\alpha} \text{Var}(E_L)$ in the simple $w = 0$ case simultaneously during the same Monte Carlo process used to compute the energy itself. We only need to sample the logarithmic derivative of the trial wavefunction alongside the local energy.

Operating in an analogous way, we can find the following gradient of the variance, needed for the more general loss function $\mathcal{L} = \langle E_L \rangle + \text{Var}(E_L)$:

$$\frac{\partial}{\partial \alpha_i} \text{Var}(E_L) = 2 \text{Cov}(O_{\alpha_i}, E_L^2) - 4 \langle E_L \rangle \text{Cov}(O_{\alpha_i}, E_L) + 2 \langle E_L \rangle \frac{\partial E_L}{\partial \alpha_i}. \quad (1.41)$$

Unfortunately, evaluating this expression numerically would require sampling many more quantities, slightly increasing computational cost, but—more relevantly—introducing more stochastic noise. To attenuate this problem, we can note that near the ground state $\langle \frac{\partial E_L}{\partial \alpha_i} \rangle \approx 0$ and thus we can claim $\langle E_L \frac{\partial E_L}{\partial \alpha_i} \rangle \approx \langle E_L \frac{\partial E_L}{\partial \alpha_i} \rangle - \langle E_L \rangle \langle \frac{\partial E_L}{\partial \alpha_i} \rangle = \langle (E_L - \langle E_L \rangle) \frac{\partial E_L}{\partial \alpha_i} \rangle$. Likewise, $E_L - \langle E_L \rangle \rightarrow 0$ as the ground state is approached. This ultimately provides the following approximation:

$$\frac{\partial}{\partial \alpha_i} \text{Var}(E_L) \approx 2 \text{Cov}(O_{\alpha_i}, E_L^2) - 4 \langle E_L \rangle \text{Cov}(O_{\alpha_i}, E_L). \quad (1.42)$$

We are now able to evaluate $\mathbf{g} = \nabla_{\alpha} \langle E_L \rangle + w \nabla_{\alpha} \text{Var}(E_L)$, although this covariance evaluation might still be susceptible to high levels of stochastic noise and therefore occasionally severely disrupt gradient-based algorithms (especially quasi-Newton methods like BFGS that approximate the Hessian matrix).

1.4.6 Nelder-Mead Algorithm

To circumvent the noise instability of the gradients, one can resort to derivative-free optimization algorithms, such as the Nelder-Mead method [13]. Relying strictly on direct point-to-point comparisons of the loss function, this algorithm operates on an observable that inherently possesses a much higher signal-to-noise ratio than its gradient.

Although Nelder-Mead is widely successful in practical applications, it is fundamentally a heuristic algorithm based on geometric transformations of a simplex. In fact, rigorous mathematical proofs of its convergence only exist for strictly convex functions in one and two dimensions, while counterexamples demonstrating non-convergence are well known for higher dimensions. Furthermore, as the simplex shrinks near the minimum, standard Nelder-Mead can still be trapped by the residual statistical fluctuations of the evaluated energy. Therefore, specific implementation strategies—which will be discussed in Chapter 3—must be adopted to dynamically mitigate the noise and ensure meaningful convergence.

1.5 Error Analysis

Once we have found the optimal parameters, we want to establish a consistent way to assign uncertainties to the energy expectations evaluated with our Monte Carlo algorithm.

We do not want to run multiple independent Monte Carlo simulations from scratch because that would be computationally prohibitive. Instead, we can extract a good estimate of the true statistical error from a single long run using resampling techniques. While methods like the *bootstrap* are conceptually simpler to implement, the *Blocking* method is specifically designed to rigorously account for the non-zero covariance of our dataset.

As previously noted, Markov chains correlate consecutive Monte Carlo steps. Consequently, the naive standard error of the mean underestimates the true statistical uncertainty. Evaluating the exact covariance requires computing the autocorrelation function, which implies a double-loop over the dataset resulting in an $\mathcal{O}(n^2)$ complexity (where $n \sim 10^6$ is the number of Monte Carlo steps); this is unfeasible. However, we can exploit the exponential decay of the autocorrelation function, which is a direct consequence of the Markov property (where the proposed step depends only on the current state).

This is the founding principle for the blocking algorithm.

1.5.1 The Automated Blocking Algorithm

Flyvbjerg and Petersen popularized the “blocking method” [4], which involves applying a “blocking transformation” to the initially correlated data points. This blocking transformation consists of grouping consecutive data points into larger “blocks” and forming a new, smaller dataset composed of the mean of each block.

For instance, let our original dataset of local energies be $X^{(0)}$ of size n . In the first blocking transformation, we average adjacent data points to create a new dataset $X^{(1)}$ of size $n/2$:

$$X_i^{(1)} = \frac{1}{2} \left(X_{2i}^{(0)} + X_{2i+1}^{(0)} \right) \quad (1.43)$$

This halving process can be repeated up to $d = \log_2(n)$ times.

As the block size increases, the new block variables $X^{(k)}$ become increasingly independent. If we compute the apparent variance of the sample mean at each blocking step k , we observe that it grows and eventually plateaus at the *true* variance of the mean, at which point the block size is larger than the autocorrelation time. Once the plateau is reached, the computed apparent variance starts oscillating significantly, mainly because the sample size has become too small and thus the statistical uncertainty of the variance itself has rapidly increased.

Marius Jonsson’s work [8] provides a rigorous mathematical proof of this method and also an automated routine to determine which blocking step is more likely to be the start of the plateau, within a set confidence level. This ultimately provides a rapid, effective and unsupervised blocking estimate.

Chapter 2

The Physical System: Neutral Atom Mixtures

2.1 The Physical System

2.1.1 Historical Research Background

Atoms of Helium-4 (^4He) have long been studied for their exceptional stability and symmetry. Typical inter-atomic interaction models include the Lennard-Jones 12-6 potential (Eq. 2.2). Historical fits of experimental measurements for the second virial coefficient (conducted on systems above 60 K) have found $\epsilon/k_B = 10.22$ K and $\sigma = 2.556$ Å [12].

The present work intends to study systems capable of modeling ^4He clusters at 0 K. We further explore the behavior of systems composed of different atomic species (or *flavors*), characterized by the same mass but a different scale of interaction.

2.1.2 Formal Definition of the System

The physical system is a mixture of N neutral atoms. All atoms are assumed to have the same mass m . Each atom is characterized by a *flavor* (i.e., the species), which can be either A or B . The number of A atoms is always equal to the number of B atoms; that is, $N_A = N_B = N/2$ and thus the total number of atoms N is always even.

The mixture is subject to two different kinds of potential: a harmonic trap and Lennard-Jones interactions. In other words, in coordinate space the physical system is described by the following Hamiltonian:

$$H = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \nabla_i^2 + \frac{1}{2}\omega^2 \sum_{i=0}^{N-1} r_i^2 + \sum_{i<j} U_{m,n}(r_{ij}), \quad (2.1)$$

where $U_{m,n}(r_{ij})$ is the Lennard-Jones interaction term

$$U_{m,n}(r_{ij}) = 4\epsilon_{m,n} \left[\left(\frac{\sigma_{m,n}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{m,n}}{r_{ij}} \right)^6 \right], \quad (2.2)$$

acting on the i -th particle, whose flavor is m , and the j -th particle, whose flavor is n . For simplicity, we set the following equalities:

$$\epsilon = \epsilon_{A,A} = \epsilon_{B,B},$$

$$\epsilon_{A,B} = \tilde{\alpha}\epsilon,$$

$$\sigma = \sigma_{A,A} = \sigma_{B,B} = \sigma_{A,B}.$$

This means pairs of particles alike in flavor $((A, A), (B, B))$ interact according to

$$U_{AA}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (2.3)$$

while pairs of particles of different flavors (A, B) interact according to

$$U_{AB}(r_{ij}) = 4\tilde{\alpha}\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right]. \quad (2.4)$$

Finally, to fix the energy and length scales of our system to the historical figures of ^4He atoms ($m = 4.0026 \text{ u}$, $\epsilon/k_B = 10.22 \text{ K}$ and $\sigma = 2.556 \text{ \AA}$, yielding $\frac{\hbar^2/(2m)}{\epsilon\sigma^2} \approx 0.09$), we set:

$$\frac{\hbar^2}{2m} = \frac{\epsilon\sigma^2}{10}, \quad (2.5)$$

and later choose $m = 1$, $\epsilon = 1$ and $\sigma = 1$ in our code.

2.2 Trial Wavefunction

To exploit the variational principle (Section 1.1), we choose the following trial wavefunction:

$$\psi(\mathbf{R}; \boldsymbol{\alpha}) = \prod_i e^{-\frac{r_i^2}{2\alpha}} \prod_{i<j} e^{-\frac{1}{2}u_\beta(r_{ij})} = \exp \left(-\frac{1}{2\alpha} \sum_i r_i^2 - \frac{1}{2} \sum_{i<j} u(r_{ij}; \beta, \gamma) \right), \quad (2.6)$$

where the Jastrow terms have the form

$$u(r_{ij}; \beta, \gamma) = \left(\frac{\beta}{r_{ij}} \right)^\gamma. \quad (2.7)$$

To avoid relying on numerical differentiation—which significantly increases the computational cost and introduces numerical artifacts—it is highly advantageous to analytically derive the quantities required for the Variational Monte Carlo implementation.

2.2.1 Local energy

The central quantity we need to evaluate is the local energy, as previously explained in Section 1.2.1. From Eq. 1.8 we have

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = \frac{1}{\psi(\mathbf{R}, \boldsymbol{\alpha})} H(\mathbf{R}) \psi(\mathbf{R}, \boldsymbol{\alpha}). \quad (2.8)$$

Plugging in the Hamiltonian's definition (Eq. 2.1), we are left with

$$E_L(\mathbf{R}, \boldsymbol{\alpha}) = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \frac{\nabla_i^2 \psi}{\psi} + \frac{1}{2} \omega^2 \sum_{i=0}^{N-1} r_i^2 + \sum_{i<j} U_{m,n}(r_{ij}). \quad (2.9)$$

While the last terms are the mere potential energy terms, the first term involves the quantity $\psi^{-1}\nabla_i^2\psi$, which can be rewritten by noticing

$$\nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2 = \frac{\nabla_i^2 \psi}{\psi}. \quad (2.10)$$

This makes $\nabla_i \ln \psi$ a central quantity to evaluate. Expressing the position vector of the i -th particle as $\mathbf{r}_i = \sum_d \hat{\mathbf{x}}_d x_{id}$ and the vector pointing from the j -th particle to the i -th particle as $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j = r_{ij} \hat{\mathbf{r}}_{ij}$, we find

$$\begin{aligned} \nabla_i \ln \psi &= \sum_d \hat{\mathbf{x}}_d \frac{\partial}{\partial x_{id}} \ln \psi = \sum_d \hat{\mathbf{x}}_d \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right] \\ &= -\frac{\mathbf{r}_i}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{\mathbf{r}_{ij}}{r_{ij}^{\gamma+2}}. \end{aligned} \quad (2.11)$$

$$\begin{aligned} |\nabla_i \ln \psi|^2 &= \sum_d \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right]^2 = \\ &= \frac{r_i^2}{\alpha^2} - \frac{\gamma\beta^\gamma}{\alpha} \sum_{j \neq i} \frac{r_i^2 - \mathbf{r}_i \cdot \mathbf{r}_j}{r_{ij}^{\gamma+2}} + \frac{\gamma^2 \beta^{2\gamma}}{4} \left[\sum_{j \neq i} \frac{1}{r_{ij}^{\gamma+1}} \right]^2 \end{aligned} \quad (2.12)$$

$$\begin{aligned} \nabla_i^2 \ln \psi &= \sum_d \frac{\partial}{\partial x_{id}} \left[-\frac{x_{id}}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{x_{id} - x_{jd}}{r_{ij}^{\gamma+2}} \right] \\ &= -\frac{D}{\alpha} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{D - 2 - \gamma}{r_{ij}^{\gamma+2}} \end{aligned} \quad (2.13)$$

Finally, we conclude the analytical expression for the local energy is given by the following equation:

$$\begin{aligned} E_L(\mathbf{R}, \alpha, \beta, \gamma) &= -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \left[-\frac{D}{\alpha} + \frac{r_i^2}{\alpha^2} + \frac{\gamma\beta^\gamma}{2} \sum_{j \neq i} \frac{D - 2 - \gamma - \alpha^{-1}r_i^2 + \alpha^{-1}\mathbf{r}_i \cdot \mathbf{r}_j}{r_{ij}^{\gamma+2}} + \right. \\ &\quad \left. + \frac{\gamma^2 \beta^{2\gamma}}{4} \sum_{j \neq i} \frac{1}{r_{ij}^{2\gamma+2}} \right] + \frac{1}{2} \omega^2 R^2 + \sum_{i < j} 4\epsilon_{m,n} \left[\left(\frac{\sigma_{m,n}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{m,n}}{r_{ij}} \right)^6 \right] \end{aligned} \quad (2.14)$$

2.2.2 Divergence cancellation

While an approximate prediction for the optimal α parameter can be obtained easily by looking at the harmonic term (the ground state of a quantum harmonic oscillator has $\sqrt{\alpha} = \sqrt{\hbar/(m\omega)}$ as its characteristic oscillation length), in order to give an estimate of the other two parameters one may study the behavior of the local energy function we have just derived explicitly in Eq. 2.14.

CHAPTER 2. THE PHYSICAL SYSTEM: NEUTRAL ATOM MIXTURES

For simplicity, first we consider a system composed of $N = 2$ particles. Looking at Eq. 2.14 and taking the limit $r_{1,2} \rightarrow 0$, we need to impose the following equality in order to suppress the higher-order energy divergences that make Eq. 2.14 far from uniform in space.

$$-\frac{\hbar^2 \gamma^2 \beta^{2\gamma}}{4m r_{1,2}^{2\gamma+2}} + 4\epsilon \tilde{\alpha} \left(\frac{\sigma}{r_{1,2}} \right)^{12} = 0 \quad (2.15)$$

This further implies

$$2\gamma + 2 = 12 \implies \gamma = 5$$

and, adhering to our convention $\frac{\hbar^2}{2m} = \frac{\epsilon \sigma^2}{10}$, we can also find a prediction for β :

$$\frac{\hbar^2}{2m} \left(\frac{25\beta^{10}}{2} \right) = \frac{\epsilon \sigma^2}{10} (40\tilde{\alpha}\sigma^{10}) \implies \beta = \left(\frac{16}{5}\tilde{\alpha} \right)^{1/10} \sigma. \quad (2.16)$$

For the symmetric case ($\tilde{\alpha} = 1$), this analytical condition predicts an optimal value of $\beta \approx 1.1233 \sigma$.

It is important to point out that in the $r_{1,2} \rightarrow 0$ limit, the $1/r^{12}$ repulsive core overwhelmingly dominates the interaction. The Jastrow factor is specifically tailored to exactly cancel this highest-order singularity. Although subleading divergences—such as the $1/r^6$ attractive potential term or the $1/r^7$ contribution arising from the Laplacian—remain technically uncanceled, they pose no analytical or numerical threat. The optimized Jastrow factor $e^{-\frac{1}{2}(\beta/r_{1,2})^5}$ drives the trial wavefunction to zero exponentially fast as particles overlap, heavily suppressing any residual lower-order polynomial singularities in the evaluation of expectation values.

Moreover, it is worth noting that this result, derived for $N = 2$, remains strictly valid for any generic N . When any two particles i and j approach each other ($r_{ij} \rightarrow 0$), their pairwise interaction completely dominates the local energy. The presence of the other $N - 2$ particles yields only finite contributions that are negligible in this limit, making the two-body divergence-canceling condition universal for the Jastrow parameters regardless of the total number of particles in the droplet.

2.2.3 Dealing with different flavors

The trial wavefunction we have used up to this point may be a good choice if $\tilde{\alpha} \approx 1$. However, if one is interested in different $\tilde{\alpha}$ values, it may be best to introduce a slightly different wavefunction:

$$\psi(\mathbf{R}; \alpha) = \prod_i e^{-\frac{r_i^2}{2\alpha}} \prod_{i < j} e^{-\frac{1}{2} \left(\frac{\beta_{m,n}}{r_{ij}} \right)^5} = \exp \left(-\frac{1}{2\alpha} \sum_i r_i^2 - \frac{1}{2} \sum_{i < j} \left(\frac{\beta_{m,n}}{r_{ij}} \right)^5 \right), \quad (2.17)$$

where m is the flavor of the i -th particle, n is the flavor of the j -th particle, $\beta_{AA} = \beta_{BB}$ and of course $\beta_{A,B} = \beta_{B,A}$. The mathematics required to utilize this wavefunction follow naturally from the previous one. Note that γ is no longer a variational parameter for the reasons discussed in Subsection 2.2.2; in fact, it is set to the constant value of $\gamma = 5$.

Other more technical aspects concerning dealing with different flavors will be discussed in Subsection 3.2.1.

2.3 Alternative methods to evaluate local kinetic energies

Throughout Subsection 2.2.1 we chose to express the kinetic term of the local energy utilizing both terms of the LHS of Eq. 2.10. An alternative option to obtain an estimate of the system's energy is to notice that the kinetic part of the Monte Carlo integral in Eq. 1.7,

$$\langle K_L \rangle = -\frac{\hbar^2}{2m} \frac{\int d\mathbf{R} |\psi|^2 \sum_{i=0}^{N-1} \frac{\nabla_i^2 \psi}{\psi}}{\int d\mathbf{R} |\psi|^2} = -\frac{\hbar^2}{2m} \frac{\int d\mathbf{R} |\psi|^2 \sum_{i=0}^{N-1} \nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2}{\int d\mathbf{R} |\psi|^2},$$

can be rewritten by exploiting the following consequences of Green's Theorem.

$$\frac{\int d\mathbf{R} |\psi|^2 \nabla_i^2 \ln \psi}{\int d\mathbf{R} |\psi|^2} = \frac{\int d\mathbf{R} |\psi|^2 \nabla_i \cdot (\nabla_i \ln \psi)}{\int d\mathbf{R} |\psi|^2} = \text{B.T.} - 2 \frac{\int d\mathbf{R} |\psi|^2 |\nabla_i \ln \psi|^2}{\int d\mathbf{R} |\psi|^2},$$

where the boundary term (B.T.) rigorously vanishes because the trial wavefunction decays exponentially at spatial infinity due to the harmonic confinement. Hence we are left with an equality between expectation values:

$$\langle \nabla_i^2 \ln \psi \rangle = -2 \langle |\nabla_i \ln \psi|^2 \rangle. \quad (2.18)$$

This allows us to find the expectation value of the local kinetic energy (and thereby of the local energy as well) by averaging over the sampled values of even just one of the two quantities $\sum_{i=0}^{N-1} \nabla_i^2 \ln \psi$ and $\sum_{i=0}^{N-1} |\nabla_i \ln \psi|^2$.

$$\langle K_L \rangle = -\frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \langle \nabla_i^2 \ln \psi + |\nabla_i \ln \psi|^2 \rangle \quad (2.19)$$

$$\langle K_L \rangle = -\frac{\hbar^2}{4m} \sum_{i=0}^{N-1} \langle \nabla_i^2 \ln \psi \rangle \quad (2.20)$$

$$\langle K_L \rangle = \frac{\hbar^2}{2m} \sum_{i=0}^{N-1} \langle |\nabla_i \ln \psi|^2 \rangle \quad (2.21)$$

Expressions 2.19, 2.20 and 2.21 are all valid in expectation. Nonetheless, their statistical variances and computational costs differ notably: evaluating Eq. 2.21 requires computing only the first-order derivatives of the wavefunction, halving the analytical operations required per step, albeit potentially at the cost of a higher variance. It is thus advisable to empirically test all three estimators to identify the most efficient trade-off.

2.4 Spatial Probability Densities

Finally, we aim to build an intuition regarding the spatial distribution of the system. More specifically, we want to answer the following questions:

1. *What is the probability density of finding **any** particle at a given point in space $\mathbf{r}$?*
2. *What is the correlation between the position of one particle and another? And what does it signify?*

2.4.1 One-Body Density

To answer the first question, we calculate the marginal probability density of a single particle by integrating the full N -body probability distribution over the position vectors of the remaining $N - 1$ particles.

Because the particles of a given flavor are indistinguishable bosons, the squared trial wavefunction $|\psi(\mathbf{R})|^2$ is symmetric under the exchange of any two particles of the same species. Furthermore, since our system is completely symmetric with respect to the two flavors ($N_A = N_B$, equal masses, and $U_{AA} = U_{BB}$), the spatial distribution of species A is physically identical to that of species B .

Consequently, computing the marginal probability density for a single generic particle and multiplying the result by the total number of particles N yields the total one-body density of the mixture, properly normalized to N :

$$\rho(\mathbf{r}) = N \frac{\int d\mathbf{r}_1 \dots d\mathbf{r}_{N-1} |\psi(\mathbf{r}, \mathbf{r}_1, \dots, \mathbf{r}_{N-1})|^2}{\int d\mathbf{R} |\psi(\mathbf{R})|^2}. \quad (2.22)$$

Since we are dealing with a many-body system, evaluating Eq. 2.22 requires a Monte Carlo approach. Conveniently, the same Metropolis algorithm devised for the VMC naturally distributes the N particles according to the target probability distribution $\propto |\psi(\mathbf{R})|^2$. This allows us to easily evaluate $\rho(\mathbf{r})$ by constructing a histogram.

For simplicity, we choose a radial histogram, meaning we divide the space into M concentric spherical shells of radii $\mathbf{r}_0, \mathbf{r}_1, \dots, \mathbf{r}_{M-1}$. At each Monte Carlo step, we measure the radial distance of every particle and increment the corresponding histogram bin. Working in D dimensions, the i -th bin counts the occurrences of particles whose positions $\mathbf{r}_j$ satisfy $\mathbf{r}_i \leq |\mathbf{r}_j| < \mathbf{r}_{i+1}$. To extract the probability $P(r)$ of finding a particle at distance r from the origin, we divide the final bin counts by the total number of Monte Carlo steps and by the bin width $\Delta \mathbf{r} = \mathbf{r}_{i+1} - \mathbf{r}_i$. Furthermore, we can find the local volumetric density $\rho(r)$, spherically averaged over the radius, by dividing the raw bin counts by the total Monte Carlo steps and by the exact geometric volume of the hyperspherical shell associated with that bin.

This same approach can be extended to concurrently obtain the one-body densities for each individual flavor group as well. Since our system is symmetric under a complete flavor exchange (i.e., all A 's swapping places with all B 's), as previously mentioned, we expect the one-body density of species A to perfectly overlap that of species B , with each contributing exactly half of the total $\rho(\mathbf{r})$.

On a final note, in order to provide a practical estimate of the statistical uncertainty for these distributions, we model the accumulation of particles in each bin as a Poisson process. Under this approximation, the standard error associated with a raw bin count C_i is given by $\sqrt{C_i}$. This absolute uncertainty is then scaled by the exact same geometric and Monte Carlo normalization factors used to compute $P(r)$ and $\rho(r)$, yielding an approximate error bar for our one-body density profiles. While the Poisson approximation provides a computationally inexpensive baseline for the statistical uncertainty, it inherently assumes independent sampling events. Because our configurations are generated via a Markov chain, consecutive particle positions are autocorrelated, meaning the true statistical error is likely larger. A mathematically rigorous estimation would require applying the blocking algorithm to each spatial bin individually; however, storing the full histogram evolution over all Monte Carlo steps imposes a severe memory overhead and a better estimate of these uncertainties is beyond the scope of this thesis.

2.4.2 Pair Correlation Function

The second question is addressed by studying the two-body radial distribution function $\rho_2(r)$ or—better—its normalization: the Pair Correlation Function $g(r) = \rho_2^{(\text{corr})}(r)/\rho_2^{(\text{uncorr})}(r)$. Theoretically, $\rho_2(r)$ represents the conditional probability density of finding a particle at a distance r from another particle. While the one-body density describes the overall macroscopic shape of the droplet relative to the center of the harmonic trap, the pair correlation function reveals the internal microscopic structure of the many-body system by measuring the spatial correlation between any two particles. In a quantum many-body system, this observable describes exactly how the presence of one atom locally modifies the spatial distribution of the surrounding ones due to their mutual interactions.

Given the binary nature of our mixture, this function naturally splits into two distinct physical quantities of interest:

- **Intra-species correlation** ($g_{\text{alike}}(r)$): describing the spatial correlation strictly between particles of the same species (i.e., A - A and B - B pairs);
- **Inter-species correlation** ($g_{\text{unlike}}(r)$): describing the spatial correlation between particles of different species (A - B pairs).

Comparing the profiles of $g_{\text{alike}}(r)$ and $g_{\text{unlike}}(r)$ provides direct quantitative evidence of the correlations between the different species present in the quantum fluid.

From a computational standpoint, $g(r)$ can be calculated by first constructing a histogram of the distances between each pair of particles, during the same Metropolis evolution needed to evaluate the one-body density. The distance between the particles at each Metropolis step already incorporates their Jastrow correlation.

This histogram can be further normalized by another histogram of inter-particle distances, this time sampling particles independently from one another by exploiting the one-body density PDF calculated earlier. Calculating the CDF of the radial one-body density histogram, one can employ the *inverse transform sampling theorem* to sample two independent radial distances r_1 and r_2 from the origin. Then, thanks to the isotropy of the system, we sample $\cos\theta$ from a uniform distribution on $[-1, 1]$, recover the relative distance via the law of cosines $|\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos\theta}$ and thus build the normalizing uncorrelated histogram.

Chapter 3

Implementation

3.1 The Code’s Objective and Structure

Our aim for this thesis is to study the neutral atom mixture system, whose background was laid out throughout Chapter 2. To accomplish this, we extended the VMC program for quantum mechanical systems, which had been previously designed and discussed in preliminary Projects [1] and [2] earlier this year¹. The Object-Oriented C++ program is meant to be efficient and modular, capable of being easily upgraded and re-utilized to study new physical scenarios or gain further insights into the ones already implemented. The output results can be graphically analyzed separately by complementary Python scripts.

The full source code is available at <https://github.com/DavideCacopardi/VMC-bachelor-thesis>. A standard `README.md` file is included and provides all necessary instructions for the user. Nevertheless, a brief overview of the program’s architecture is provided below to facilitate the comprehension of the simulation workflow and highlight key computational strategies.

3.1.1 Summary of the Main Routines

The execution of the program is subdivided into four routines:

1. *Optimization*: Starting from an initial parameter guess, the selected automatic optimization algorithm attempts to find the parameter vector which minimizes the set loss function $\mathcal{L} = E + w \text{Var}(E)$.
2. *FinalMC & Blocking*: A high-precision Monte Carlo energy estimation is performed² and the results are post-processed using the *Blocking* resampling technique.
3. *Spatial Distribution*: A One-Body Density histogram and a Pair Correlation histogram are output; additionally, the spatial trajectories of the particles during the Metropolis evolution are logged².

¹Note that Project [2] dealt with the use of neural networks as the system’s wavefunction. Here we shall not delve into those techniques; they are, however, still available as options in the present version of the code.

²Ordinarily the optimal parameters generated by the *Optimization* step are used, but this can be overridden to allow for a manual parameter choice.

4. *Mesh Grid*: A linearly equally-spaced parameter grid is generated and a Monte Carlo energy estimation is run at each node. This is primarily aimed at gaining an intuitive understanding of the hyper-surface determined by the energy function $E(\alpha)$.

These routines can be selectively enabled via command-line interface (CLI) toggles, and will be executed in the aforementioned sequential order. All tunable parameters—encompassing both physical variables and technical settings—can be configured in the `config.json` file.

3.1.2 The Idea behind the Implemented Classes

The `System` class represents the complete quantum system. It acts as a bridge between the respective `Hamiltonian`, `WaveFunction`, `Particle` and `Solver` objects. It coordinates the Monte Carlo processes and returns the requested `Sampler` objects.

The `Hamiltonian` classes define the chosen physical environment interface and compute the Local Energy accordingly.

The `WaveFunction` classes define the chosen trial wavefunction and its optimizable parameters; analytical overrides for the relevant derivatives can be implemented here.

The `Particle` class defines the attributes (such as position and *flavor*) of each of the particles composing the physical system.

The `Solver` classes define the Metropolis algorithm employed to construct the Monte Carlo Markov Chain; more generally, they provide the step function called by the `System`'s methods.

The `Sampler` classes define what is to be sampled at each Monte Carlo step and handle the associated statistics; as previously mentioned, they are output by the `System`'s methods as they retain all relevant simulation results, in addition to optionally logging them to file.

The `MCEngine` class wraps the Monte Carlo methods of the `System` class and handles the multi-threading execution.

The `VMCOptimizer` classes implement optimization algorithms manually or wrap imported ones from open-source libraries.

Other more technical classes and source files are well documented in the GitHub repository.

3.2 Implementation Choices

In this section we explain the most notable algorithmic implementations from a computational standpoint.

3.2.1 Dealing with Different Flavors and Automating the Solvers

The `setupRandomUniformInitialState` function initializes the particles of the system. Traditionally, this solely implies generating random coordinates from a user-defined uniform distribution and optionally enforcing additional constraints such as setting all particles at a safe distance from each other (if dealing with strongly repulsive potentials). In our case we also need to assign a different *flavor* to each particle. We chose to assign flavor *A* to all evenly indexed particles and flavor *B* to the remaining half. One might alternatively assign a random flavor to each particle, relaxing the $N_A = N_B = N/2$ assumption; however, this is better suited for a larger number of particles and lies beyond the scope of this thesis.

Furthermore, since our system’s observables depend on the particles’ flavors, special care must be taken regarding the Monte Carlo evolution of the particles. While bosonic symmetry is naturally respected by the standard Metropolis Markov chain, isolated MCMC samplers can easily get trapped in metastable regions of the configuration space, leading to highly correlated samples and erratic energy estimates [16]. To ensure proper ergodicity and species mixing, we find it necessary to forcefully introduce a *particle swap* approximately and arbitrarily every ~ 10 *step* proposals of the Markov Chain. This is implemented as a 1/10 stochastic flavor swap, utilizing the same acceptance criterion employed in Brute Force Metropolis (Subsection 1.3.2) because the flavor-swap proposal is perfectly symmetric.

Finally, we implemented a smart `stepLength/timeStep` auto-tuning system. It is a reasonable rule of thumb to try to set the `stepLength` parameter of a Brute Force Metropolis to a value yielding an acceptance ratio of approximately $\sim 50\%$. Likewise, it is advisable to set the `timeStep` parameter of an Importance Sampling Metropolis to target an acceptance ratio of roughly $\sim 90\%$. These optimal step values depend considerably on the physical system and configurations being analyzed; by default, this forces the user to manually restart the VMC code several times to find a suitable parameter choice. To circumvent this, we implemented an automatic `step`-parameter tuning, acting every ~ 100 equilibration Metropolis steps, applying a simple 10% increase or decrease based on the acceptance ratio measured during the preceding ~ 100 steps. Note that this tuning must occur strictly during the equilibration phase, because changing the `step` parameter in the middle of a Metropolis sampling process compromises the detailed balance, thereby possibly altering the accuracy of the sampling itself; this bias is not notably impactful for the simple purpose of the equilibration steps.

3.2.2 Optimization of Parameters

Three main optimization algorithms are implemented: BFGS, Adam and Nelder-Mead. While BFGS and Nelder-Mead implementations are wrappers around the NLOpt open-source library [7], the Adam algorithm was implemented manually³.

Automated Number of Monte Carlo Cycles Increase

Since precise Monte Carlo runs can take a considerable amount of time, we implemented the capability to start our optimization algorithms with a relatively low number of Monte Carlo steps in order to quickly reach the neighborhood of the minimum. Only subsequently are we interested in refining the search with more Monte Carlo cycles, especially when the energy well is not steep. All three algorithms feature an automated increase of Monte Carlo cycles. Every `autoMCincr_wait` = 10 optimization steps, the following signal-to-noise ratio condition is checked.

$$|\mathcal{L}_i - \mathcal{L}_{i-1}| < 4\sigma_E \quad (3.1)$$

If it evaluates to `true`, the number of Monte Carlo cycles is increased by 20% and the check is repeated. This increase can be triggered for a maximum of `c_max_improvement_tries` = 5 consecutive times. Please note that this automated increase system is a ‘black-box’ to the optimization algorithm. That is, the optimization algorithm only sees the most recent evaluation of $\mathcal{L}_i$ in order not to compromise the validity of more than one optimization step.

³Adam is also the optimization algorithm of choice to train the neural networks. For that specific purpose, the PyTorch library [14] and environment were utilized; however, as previously stated, this falls outside the scope of the present thesis.

The loss function $\mathcal{L} = E + w \text{Var}(E)$ —the driver of the optimization algorithm—depends on good energy and gradient estimates, and possibly even on the evaluated energy variance and its gradient in turn, too. Growing the number of Monte Carlo cycles throughout the process can abruptly reduce the stochastic noise, making the newly evaluated $\mathcal{L}_i$ appear deceptively lower than $\mathcal{L}_{i-1}$, potentially stalling the algorithm if not handled properly. Finally, it is worth noting that the RHS of Ineq. 3.1 is a very rough estimate of the uncertainty on $\mathcal{L}_i$. A rigorous evaluation would require not only sampling the error on $\text{Var}(E)$, but also running a resampling technique (such as the *Blocking* algorithm) to account for the inherent covariance of the Markov Chain. We opt to bypass such high-precision calculations here, as this automated step-increase machinery merely aims to accelerate a process that would otherwise require manual trial-and-error.

Stopping Criteria

Both `NLopt` wrappers use `set_xtol_rel` as their primary stopping criterion (usually set to $\sim 10^{-5}$); this means that the optimization process terminates when the length of the parameter step $(\Delta\alpha)_i = \alpha_i - \alpha_{i-1}$ is such that $(\Delta\alpha)_i / \alpha_i < \text{xtol_rel}$.

Additional fallback `NLopt` stopping criteria are enforced via `set_maxeval(400)` and `set_maxtime(10800)`.

Conversely, the Adam optimizer always executes a user-defined number `Adam_nSteps` of optimization steps.

On top of all this, the `SIGINT` ‘external interrupt’ listener is redirected to gracefully terminate the optimization process when it is still underway, enabling the user to skip the remaining optimization steps and proceed to the next routines without losing progress.

3.2.3 Learning Rate Reduction on Plateau

Although Adam is an adaptive learning rate scheme in itself, we implemented a supplementary loss-based learning-rate adjuster, in order to force a more precise convergence when a plateau is reached.

If the condition

$$\mathcal{L}_i < \left(\min_{j \leq i} \mathcal{L}_j \right) - \text{Adam_min_improvement}$$

evaluates to `false` for `Adam_max_patience` consecutive optimization steps, the learning rate is divided by 2 and the `min_improvement` parameter by 10.

3.2.4 Multi-threading & Blocking

As anticipated in Subsections 3.1.1 and 3.1.2, the code is designed to execute parallel Monte Carlo samplings across a user-defined number of threads (`nThreads`). The resulting `Sampler` objects are then averaged together into a single comprehensive output `Sampler`. Running multiple independent MCMC samplers in parallel and combining their samples is a primary strategy not only to speed up computation, but to directly reduce the final statistical variance of the estimators [16].

As for the Blocking resampling step, our code utilizes Jonsson’s C++ implementation, available at <https://github.com/computative/block>. There, the confidence level is set to 5%. This algorithm requires the total dataset size to be a power of 2, as explained in Subsection 1.5.1. The number of Monte Carlo cycles for the *FinalMC* and *Mesh Grid* runs can indeed be specified by the user as any power of 2. However, when utilizing multiple threads, the

actual number of Monte Carlo steps executed by each individual thread is a $1/\mathbf{nThreads}$ fraction of the total. Since the Blocking algorithm must individually resample each contiguous batch of data produced by each thread, the number of Monte Carlo cycles run by each thread must also be a power of 2. Consequently, if one wishes to obtain a Blocking uncertainty estimation, **nThreads** must be chosen as a power of 2 itself.

Chapter 4

Execution, Data and Results

4.1 Parameter Choice and Analytical Predictions

To benchmark our Variational Monte Carlo code and investigate the physical properties of the neutral atom mixture, we define a consistent set of algorithmic configurations and physical parameters and derive key analytical reference values to guide the analysis of our numerical results.

Unless otherwise specified, all simulations employ Importance Sampling Metropolis Monte Carlo stabilized by the Umrigar drift (Subsection 1.3.4) with interpolation parameter $a = 1$, alongside the stochastic particle-flavor swap routine (discussed in Subsection 3.2.1). We study three-dimensional systems ($D = 3$) composed of $N = 2, 4, 6, 8$ (and occasionally 12) particles of identical unit mass $m = 1$, governed by Hamiltonians configured as follows.

4.1.1 Lennard-Jones Potential

For the Lennard-Jones potential, we adopt standard unit parameters: $\sigma = 1$, $\epsilon = 1$. To satisfy our energy-scale convention $\frac{\hbar^2}{2m} = \frac{\epsilon\sigma^2}{10}$, the kinetic diffusion constant is fixed to $D = \frac{\hbar^2}{2m} = 0.1$. This sets the de Boer parameter to $\Lambda^2 = \hbar^2/(m\sigma^2\epsilon) = 0.2$ (slightly higher than the typical $\Lambda^2 = 0.18$ of ^4He). To explore different interaction regimes, we analyze three values of the cross-interaction factor: $\tilde{\alpha} = 1.00, 0.40, 0.05$.

Setting the derivative of a Lennard-Jones interaction term to zero, one can find its minimum:

$$r_{ij}^{(\min)} = 2^{1/6}\sigma = 2^{1/6} \approx 1.122, \quad (4.1)$$

$$\begin{aligned} U_{AA}(r_{ij}^{(\min)}) &= -\epsilon = -1, \\ U_{AB}(r_{ij}^{(\min)}) &= -\tilde{\alpha}\epsilon = -\tilde{\alpha}. \end{aligned} \quad (4.2)$$

4.1.2 Harmonic Potential

As for the harmonic trap potential, we select two distinct angular frequencies ω , reported in Table 4.1.

The ground state of a 3D quantum harmonic oscillator has energy $E_0 = \frac{3}{2}\hbar\omega$ and a characteristic Gaussian width $\sqrt{\alpha} = \sqrt{\frac{\hbar}{m\omega}}$. The energy of a 3D system of N particles is thus

$E_N = \frac{3}{2}N\hbar\omega$. The α and E_0 values corresponding to the chosen ω values are also defined and listed in Table 4.1.

ω	α	E_0
$\omega_1 := 0.111803$	4	0.074 999 732 4
$\omega_2 := 0.027951$	16	0.018 750 100 8

TABLE 4.1: Quantum harmonic potential predictions for parameter α and ground-state energies E_0 .

4.1.3 Two-body Systems

We now consider the simple case $N = 2$ and try to solve it analytically in order to benchmark our $N = 2$ interacting results.

The two-body Hamiltonian

$$H = -\frac{\hbar^2}{2m}\nabla_1^2 - \frac{\hbar^2}{2m}\nabla_2^2 + \frac{1}{2}m\omega^2(r_1^2 + r_2^2) + U_{1,2}(r_{1,2})$$

can be rewritten in the following form:

$$H = -\frac{\hbar^2}{2(2m)}\nabla_{\text{CM}}^2 + \frac{1}{2}(2m)\omega^2 r_{\text{CM}}^2 - \frac{\hbar^2}{2(m/2)}\nabla_{1,2}^2 + \frac{1}{2}(m/2)\omega^2 r_{1,2}^2 + U_{1,2}(r_{1,2}), \quad (4.3)$$

where $\mathbf{r}_{\text{CM}} = (\mathbf{r}_1 + \mathbf{r}_2)/2$ is the center of mass and $\mathbf{r}_{1,2} = \mathbf{r}_1 - \mathbf{r}_2$ is the usual relative position vector. The ground-state energy E of the total system is then given by the sum of the ground-state energy of the system of the center of mass E_{CM} , governed by the harmonic oscillator Hamiltonian

$$H_{\text{CM}} = -\frac{\hbar^2}{2(2m)}\nabla_{\text{CM}}^2 + \frac{1}{2}(2m)\omega^2 r_{\text{CM}}^2,$$

and the ground-state energy of the system of the relative coordinates $E_{1,2}$, governed by the more complex

$$H_{1,2} = -\frac{\hbar^2}{2(m/2)}\nabla_{1,2}^2 + \frac{1}{2}(m/2)\omega^2 r_{1,2}^2 + U_{1,2}(r_{1,2}).$$

The ground state of the center of mass system is thus the usual one of a 3D quantum harmonic oscillator: $E_{\text{CM}} = \frac{3}{2}\hbar\omega = E_0$. Conversely, for the relative coordinates' system, we need to solve the following radial eigenvalue equation:

$$H_{1,2}R(r_{1,2}) = E_{1,2}R(r_{1,2}).$$

Dropping the subscript on the variable and solving for the usual s -wave $u(r) = rR(r)$, with $u(0) = 0$, one obtains:

$$-\frac{\hbar^2}{2(m/2)}\frac{d^2u}{dr^2} + \left[\frac{1}{2}(m/2)\omega^2 r^2 + 4\epsilon\tilde{\alpha} \left(\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right) \right] u(r) = Eu(r).$$

Unfortunately, this equation is not analytically solvable, but we can obtain a numerical solution by discretizing the second derivative and the domain of the r variable to $\{r_{\min}, r_{\min} +$

$\delta r, r_{\min} + 2\delta r, \dots, r_{\max} - \delta r, r_{\max}\}$ and then diagonalizing the resulting tri-diagonal matrix:

$$\begin{bmatrix} D(r_{\min}) & d & 0 & \dots & 0 \\ d & D(r_{\min} + \delta r) & d & \dots & \vdots \\ 0 & d & D(r_{\min} + 2\delta r) & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & d \\ 0 & \dots & 0 & d & D(r_{\max}) \end{bmatrix},$$

where we defined

$$d = -\frac{1}{(\delta r)^2} \frac{\hbar^2}{2(m/2)}$$

and

$$D(r) = \frac{1}{(\delta r)^2} \frac{\hbar^2}{(m/2)} + \frac{1}{2}(m/2)\omega^2 r^2 + 4\epsilon\tilde{\alpha} \left(\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right).$$

To do this, we write a simple Python script exploiting the `linalg.eigh_tridiagonal` function of the well-known Scipy library. This function internally calls `stebz` from the LAPACK library, which—in turn—computes the selected eigenvalues of a real symmetric tridiagonal matrix using bisection (more info at docs.scipy.org).

We choose $r_{\min} = 10^{-3}$, $n = 10^5$ and $r_{\max} = 50$, and obtain the estimates reported in Table 4.2. Notice how the $E_{1,2}$ estimates for the non-interacting ($N = 2; \tilde{\alpha} = 0$) systems agree with the completely analytical $E_{\text{CM}} = E_0$ within the selected number of significant figures.

ω	$\tilde{\alpha}$	E_{CM}	$E_{1,2}$	E_{tot}
ω_1	1.00	0.074 999 732 4	0.0537	0.1287
ω_1	0.40	0.074 999 732 4	0.0813	0.1563
ω_1	0.05	0.074 999 732 4	0.0881	0.1632
ω_1	0.00	0.074 999 732 4	0.0750	0.1500
ω_2	1.00	0.018 750 100 8	0.012 16	0.030 91
ω_2	0.40	0.018 750 100 8	0.019 17	0.037 92
ω_2	0.05	0.018 750 100 8	0.020 30	0.039 06
ω_2	0.00	0.018 750 100 8	0.018 75	0.037 50

TABLE 4.2: Predictions for the ground-state energy of the two-body systems. E_{CM} is an analytical estimate (the same of Table 4.1); $E_{1,2}$ is obtained by discretization of domain and automatic diagonalization.

4.2 Non-Interacting System

In order to test the efficacy of our code, we first disable the interaction potential and benchmark the computed energies against the analytical solutions for independent harmonic oscillators.

The optimization algorithms require the initial guess to be set within the same strictly convex region as the sought-after minimum and may easily fail due to the considerable stochastic noise imposed by non-optimal parameter neighborhoods. Therefore, from now on, we choose to run the implemented Mesh Grid procedure as a way to double check the optimization results; in fact, the equally-spaced manual grid-search provides a safe rough estimate of the optimal set of parameters. If the automated optimization algorithm fails to converge to the same minimum found by the Mesh Grid method (within a margin of error), then one can repeat the optimization process employing a different algorithm or tailoring the optimization parameters to the observed behavior.

N	ω	$E^{(\text{exp.})} = NE_0$	optimizer	w	$\alpha^{(0)}$	α	E
2	ω_1	0.149 999 464 9	Nelder-Mead	0.05	4	4.0023	0.149 999 50(1)
2	ω_1	0.149 999 464 9	Nelder-Mead	0.05	4	3.9976	0.149 999 48(2)
2	ω_1	0.149 999 464 9	BFGS	0.05	4	4.0000	0.149 999 464 9(1)
2	ω_1	0.149 999 464 9	Mesh Grid	0.00	—	4.0(4)	0.149 999 464 0(4)
4	ω_1	0.299 998 929 7	BFGS	0.05	4	4.0000	0.299 998 929 8(2)
4	ω_1	0.299 998 929 7	Mesh Grid	0.00	—	4.0(4)	0.299 998 928 9(8)
6	ω_1	0.449 998 394 6	BFGS	0.05	4	4.0000	0.449 998 394 4(3)
6	ω_1	0.449 998 394 6	Mesh Grid	0.00	—	4.0(4)	0.449 998 395(1)
8	ω_1	0.599 997 859 4	BFGS	0.05	4	4.0000	0.599 997 859 4(4)
8	ω_1	0.599 997 859 4	Mesh Grid	0.00	—	4.0(4)	0.599 997 860(1)
2	ω_2	0.037 500 201 6	BFGS	0.05	12	16.000	0.037 500 201 63(7)
2	ω_2	0.037 500 201 6	Mesh Grid	0.00	—	16.0(4)	0.037 500 201 5(2)
4	ω_2	0.075 000 403 2	BFGS	0.05	16	16.000	0.075 000 403 1(1)
4	ω_2	0.075 000 403 2	Mesh Grid	0.00	—	16.0(4)	0.075 000 403 3(3)
6	ω_2	0.112 500 604 9	BFGS	0.05	16	16.000	0.112 500 605 0(1)
6	ω_2	0.112 500 604 9	Mesh Grid	0.00	—	16.0(5)	0.112 500 604 7(5)
8	ω_2	0.150 000 806 5	BFGS	0.05	16	16.000	0.150 000 806 4(2)
8	ω_2	0.150 000 806 5	Mesh Grid	0.00	—	16.0(5)	0.150 000 806 5(7)

TABLE 4.3: VMC results for non-interacting harmonic $V = \frac{1}{2}\omega^2 R^2$ systems. Different optimization methods were tested. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Both optimization algorithms used `x.tol` = 10^{-5} as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.

The results are shown in Table 4.3 and perfectly agree with the analytical predictions¹. Notice how the BFGS optimization converges to the expected result with remarkable accuracy, even when initialized far from the optimal value (e.g., $\alpha^{(0)} = 12$). Nelder-Mead, on the other hand, exhibits minor residual deviations on the order of $\sim 10^{-3}$, primarily arising from simplex stagnation in the presence of stochastic noise.

¹Further analysis on non-interacting systems utilizing a previous version of this program were conducted in Project [1].

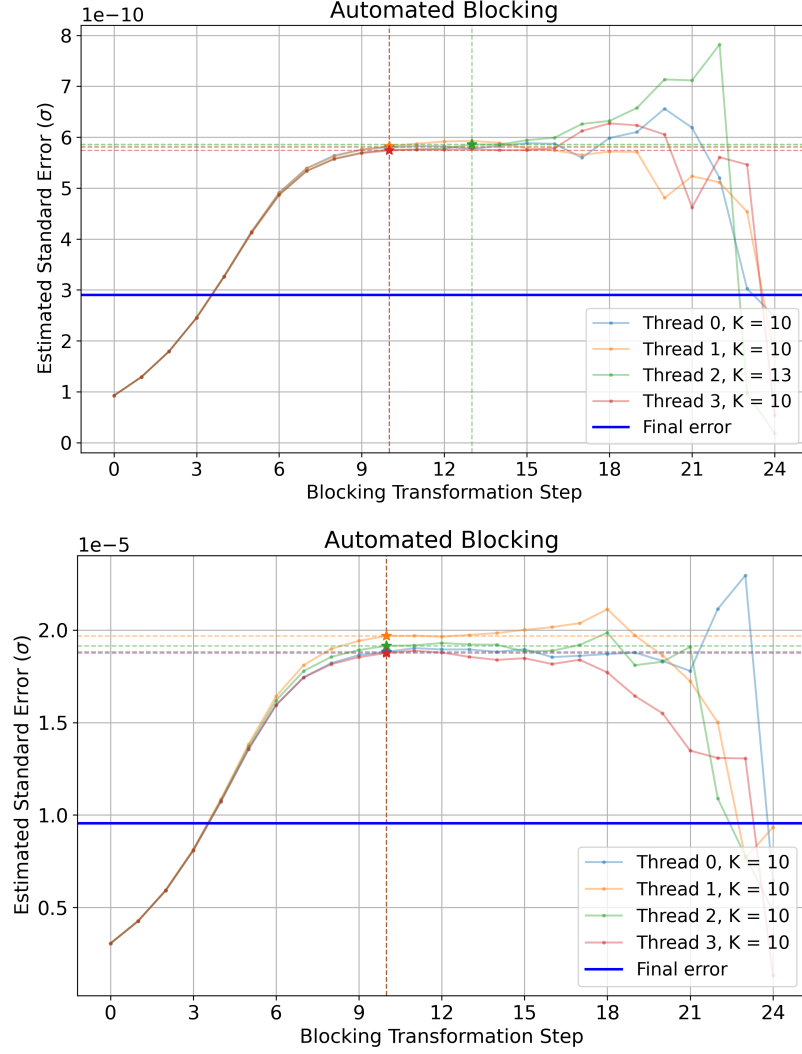


FIGURE 4.1: Estimated standard error vs number of blocking transformations via the automated blocking algorithm (Subsection 3.2.4). The first plot analyzes the final MC (2^{27} MC cycles) run for the $N = 6$, ω_1 non-interacting system (Table 4.3); the second plot refers to a final MC run for the same system but manually setting a non-optimal $\alpha = 4.5$, resulting in an energy estimate of 0.45311 ± 0.00001 as opposed to the naive pre-blocking result 0.453107 ± 0.000002 .

Figure 4.1 shows how Marius Jonsson’s *Blocking* algorithm successfully detects the plateau of the apparent variance trend in each data-set. On the other hand, the uncertainty associated with the optimal parameters obtained from automated optimization algorithm is harder to estimate: several factors should be taken into account, including the energy gradients evaluated at convergence and the stochastic noise that affects them. Because of this henceforth we choose to list the found optimal parameters up to 5 significant digits (since we usually set $\mathbf{x.tol} = 10^{-5}$), but more conservative and realistic estimates would probably stop at the 3rd significant digit, assessing the qualitative flatness of the energy around the minimum and the stochastic errors during optimization. Mesh Grid optimal parameters are reported with uncertainty equal to the grid’s spacing width.

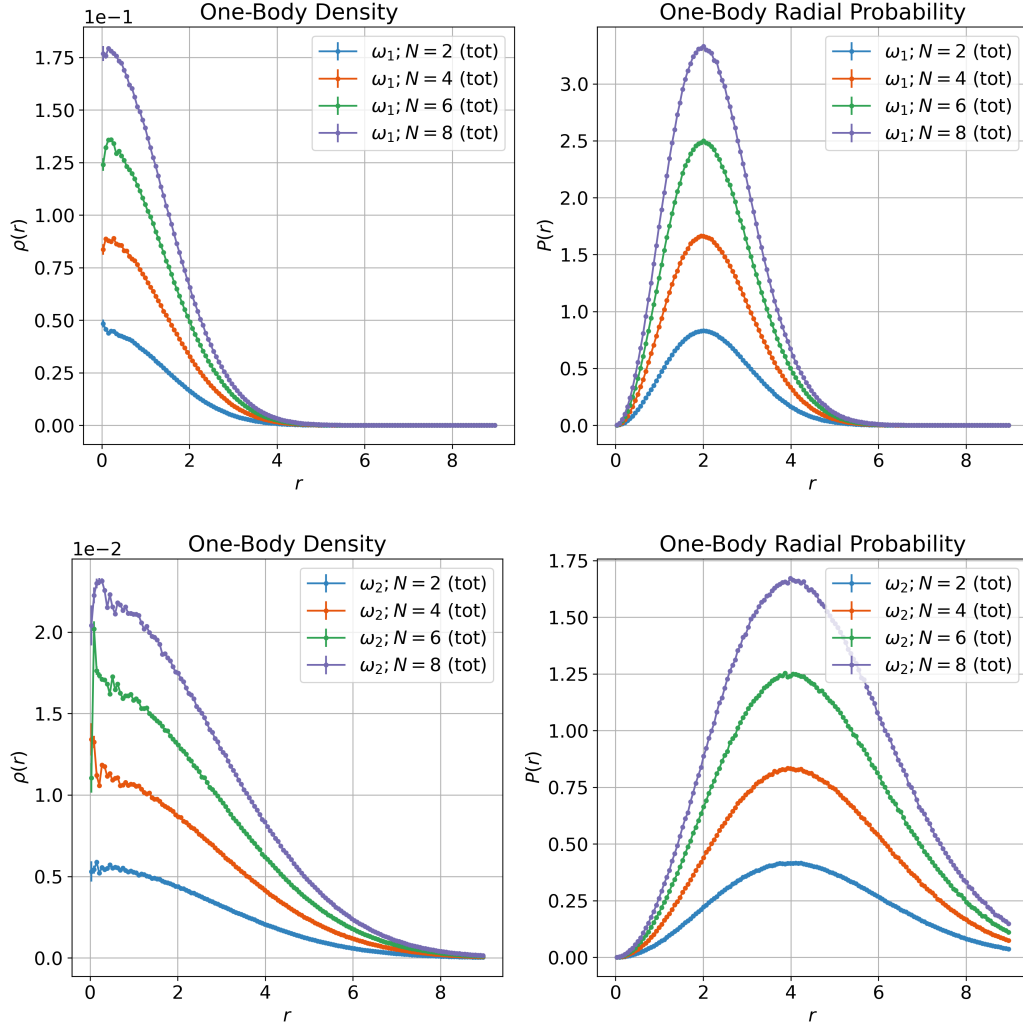


FIGURE 4.2: Radial one-body densities ($\sim 10^7$ MC steps) of the non-interacting systems. As expected, these non-normalized plots scale with the number of particles and perfectly match the Gaussian widths $\sqrt{\alpha} = 2, 4$.

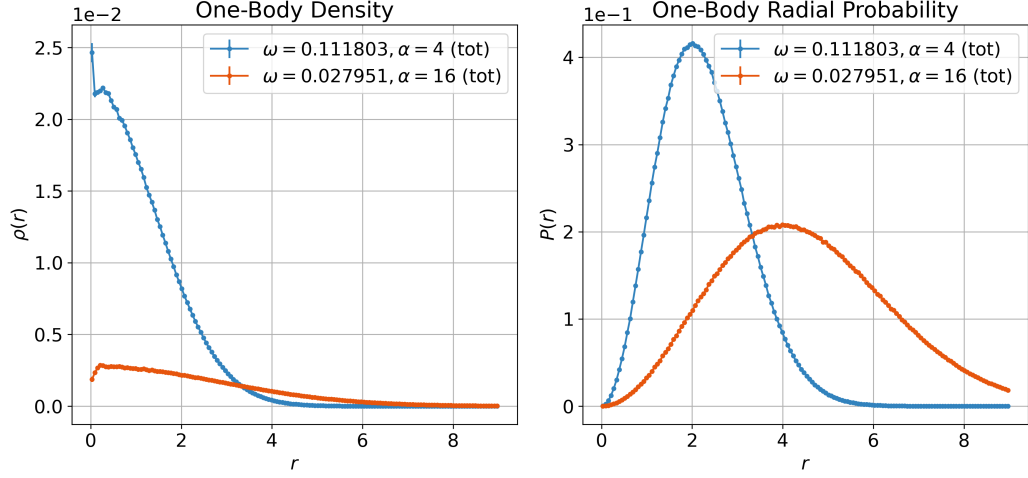


FIGURE 4.3: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 4$ non-interacting systems. The plot is normalized by the number of particles. The stronger the confinement ω , the tighter the distribution.

Finally, comparative tests on the local kinetic energy estimators confirmed that Eq. 2.20 yields a lower statistical variance than Eq. 2.21, while the complete laplacian formulation of Eq. 2.19 consistently exhibits the minimal variance overall. An explicit comparison is later shown in Figure 4.15 for an interacting system. Consequently, Eq. 2.19 is adopted as the primary kinetic estimator throughout all simulations.

4.3 Interacting Systems

4.3.1 Strong confinement

Next, we investigate the fully interacting Lennard-Jones mixture under harmonic confinement.

We first consider the strong confinement regime $\omega = \omega_1$. The results are shown in Table 4.4 and the most relevant ones are visualized in Figure 4.4; some one-body density plots and pair correlation plots are presented in Figures 4.8 to 4.13.

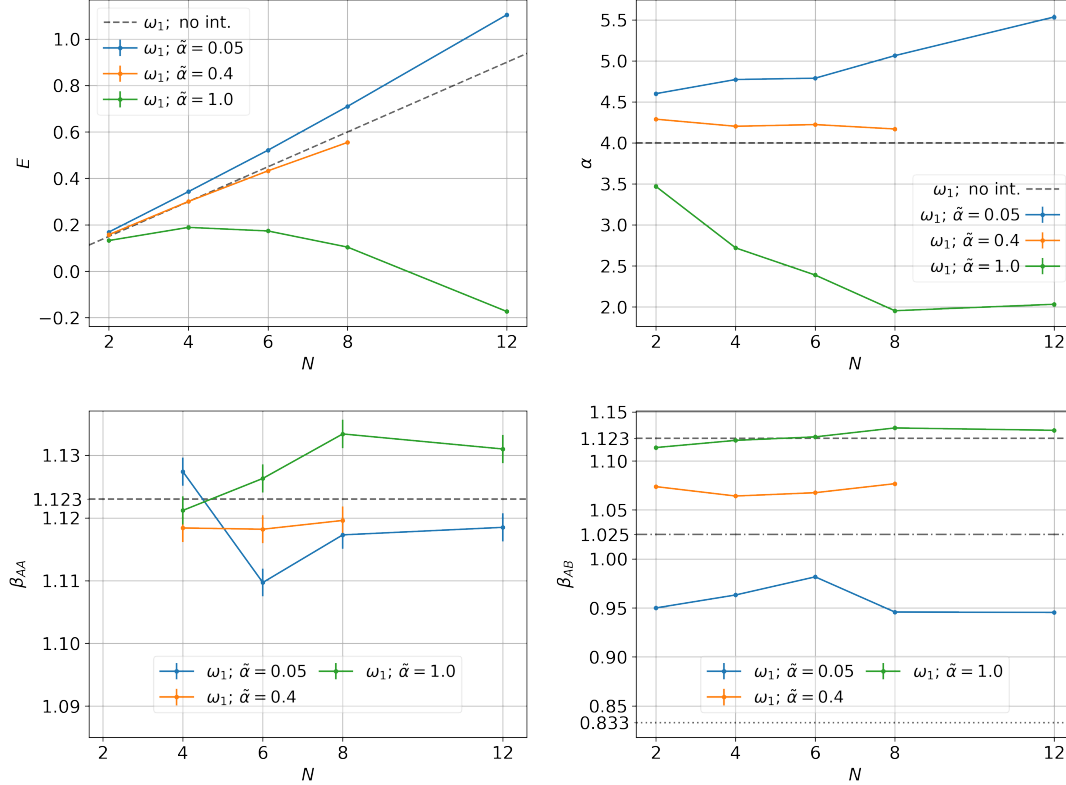


FIGURE 4.4: Best energy, α and β values obtained for the interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement. The optimal parameters are displayed with a roughly estimated 0.2% relative error because of the reasons discussed in Section 4.2

Disclaimer about the encountered limitations of the wavefunction and of the optimizers

All measured energies are determined with high statistical precision, with blocking-estimated relative errors ($\lesssim 0.01\%$) (Table 4.4). This confirms that the Monte Carlo sampling itself is well-converged for each fixed parameter set, but it does not say how close the optimized trial wavefunction is to the true ground state.

That question is better addressed by two preliminary considerations: first, we compare the results we obtained for the $N = 2$ systems to the predictions we made in Subsection 4.1.3 (i.e., compare the $E^{(\text{exp.})}$ to the E in Table 4.4). All computed energies overestimate their expected value (+2.8% in the $\tilde{\alpha} = 1.00$ case, +0.8% in the $\tilde{\alpha} = 0.40$ case and +3.4% in the

TABLE 4.4: VMC results for interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement. Different optimization methods were tested ("N-M" stands for "Nelder-Mead"; "M-G" stands for "Mesh Grid"). The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05, 0.05, 0.05]$ as initial simplex size and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Adam ran 120 steps with 10^7 MC cycles and an initial learning rate of 0.03. Nelder-Mead and BFGS used $\mathbf{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values.

N	$\tilde{\alpha}$	$E^{(\text{exp.})}$	Optim.	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
2	1.00	0.1287	N-M	0.05	4	—	1	3.5043	—	1.1169	0.132 306(8)
2	1.00	0.1287	BFGS	0.05	3.5	—	1.1	3.4999	—	1.1177	0.132 305(8)
2	1.00	0.1287	Adam ^{††}	0.0005	4	—	0.95	3.4676	—	1.1137	0.132 280(7)
2	1.00	0.1287	M-G	0.00	—	—	—	3.6(2)	—	1.08(7)	0.132 24(2)
4	1.00	—	N-M	0.05	2.3	1	1	2.6703	1.1227	1.1240	0.189 19(3)
4	1.00	—	Adam	0.05	2.3	1	1	2.7472	1.1193	1.1180	0.189 04(3)
4	1.00	—	N-M	0.05	2.3	1	1	2.7195	1.1212	1.1210	0.189 03(3)
4	1.00	—	M-G	0.00	—	—	—	2.6(3)	1.12(4)	1.12(4)	0.1883(3)
6	1.00	—	N-M	0.05	2.3	1	1	2.3881	1.1263	1.1246	0.173 59(7)
6	1.00	—	M-G	0.00	—	—	—	2.2(2)	1.15(7)	1.15(7)	0.1830(3)
8	1.00	—	N-M	0.05	2.3	1	1	1.9532	1.1334	1.1337	0.1041(1)
8	1.00	—	M-G	0.00	—	—	—	2.0(2)	1.15(7)	1.15(7)	0.1121(6)
12*	1.00	—	Adam [‡]	0.0005	2	1.1	1.1	2.0320	1.1310	1.1313	-0.1731(2)
2	0.40	0.1563	N-M	0.05	4.3	—	1	4.2884	—	1.0738	0.157 485(8)
2	0.40	0.1563	M-G	0.00	—	—	—	4.2(3)	—	1.08(7)	0.157 50(4)
4	0.40	—	N-M	0.05	4.3	1	1	4.2020	1.1184	1.0642	0.300 43(3)
4	0.40	—	M-G	0.00	—	—	—	4.2(3)	1.12(4)	1.08(4)	0.2997(1)
6	0.40	—	N-M	0.05	4.3	1	1	4.2228	1.1182	1.0676	0.432 46(5)
6	0.40	—	M-G	0.00	—	—	—	3.9(3)	1.08(7)	1.08(7)	0.4337(5)
8	0.40	—	N-M	0.05	4.3	1	1	4.1691	1.1196	1.0768	0.5549(1)
8	0.40	—	M-G	0.00	—	—	—	3.9(3)	1.15(7)	1.08(7)	0.5599(3)
2	0.05	0.1632	N-M	0.05	4.5	—	1	4.4407	—	0.9423	0.168 98(2)
2	0.05	0.1632	N-M	0.05	4.6	—	1	4.6000	—	0.9500	0.168 71(2)
2	0.05	0.1632	M-G	0.00	—	—	—	4.6(2)	—	1.01(7)	0.168 63(9)
4	0.05	—	N-M	0.05	4.4	1	1	4.3110	1.1169	0.9346	0.345 22(6)
4	0.05	—	N-M	0.05	4.6	1	1	4.5084	1.1169	0.9410	0.343 60(6)
4	0.05	—	N-M	0.05	4.7 [†]	1.1	1	4.7722	1.1274	0.9632	0.342 78(6)
4	0.05	—	M-G	0.00	—	—	—	4.8(4)	1.12(4)	1.00(4)	0.3425(4)
6	0.05	—	N-M	0.05	4.6	1	1	4.4306	1.1188	0.9401	0.5259(1)
6	0.05	—	N-M	0.05	4.8 [†]	1.1	1	5.1093	1.1261	0.9334	0.5259(1)
6	0.05	—	Adam [‡]	0.0005	4.8	1.1	1	4.7884	1.1097	0.9817	0.5213(1)
6	0.05	—	M-G	0.00	—	—	—	4.8(4)	1.16(12)	1.04(12)	0.5304(4)
8	0.05	—	N-M	0.05	4.6	1	1	4.8579	1.1174	0.9481	0.7094(2)
8	0.05	—	N-M	0.05	4.8 [†]	1.1	1	5.0631	1.1173	0.9458	0.7101(2)
8	0.05	—	M-G	0.00	—	—	—	4.8(4)	1.16(12)	1.04(12)	0.7226(6)
12*	0.05	—	N-M	0.05	4.8	1.11	0.83	5.5925	1.1202	0.9372	1.1109(3)
12*	0.05	—	N-M	0.05	5.3	1.1	0.94	5.3442	1.1185	0.9454	1.1044(3)
12*	0.05	—	M-G	0.00	—	—	—	5.0(5)	1.112**	0.95(7)	1.098(2)

* Bonus run.

[†] Nelder-Mead ran with $\pm [0.5, 0.05, 0.05]$ as initial state setting.

^{††} Adam ran 375 steps with 10^6 MC cycles, an initial learning rate of 0.05 and `autoMCincr_wait` = 100.

[‡] Adam ran 1000 steps with 5×10^5 MC cycles, an initial learning rate of 0.05 and `autoMCincr_wait` = 500.

** $\beta_{AA} = 1.112$ was held constant throughout the grid-search (i.e., the parameter was not optimized).

$\tilde{\alpha} = 0.05$ case). On one hand, this proves that the program itself is likely working correctly (only upper bounds to the predictions are found); on the other, the discrepancies are large enough to raise the suspicion that the employed wavefunction is not capable of accurately representing the true ground state.

A second consideration comes from the Vanishing-Variance Principle (Subsection 1.2.2): if ψ were close to the ground state, the variance landscape and the energy landscape should share the same minimum. As Figures 4.5 and 4.6 show, this is only approximately true—the variance valley is often visibly offset from the energy valley—reaffirming that the employed Jastrow ansatz captures the dominant physics but is not a high-accuracy representation of the true ground state.

We return to this limitation and its likely origin subsequently in this subsection.

Furthermore, after a few failed attempts at employing gradient-based optimizers, we had to resort to Nelder-Mead in spite of its well-known drawbacks. The gradient of the variance had often overwhelmed that of the energy, generally leading both the α and β parameters to higher values seeking to disperse the bosons and thereby prevent sampling points close to the singularities, even when it would actually be energetically favorable to do so. We should also remind ourselves that the gradient of the variance is approximated as derived in Subsection 1.4.5 and is therefore a valid estimate solely around the true ground-state minimum. Only at the end did we manage to find better results by lowering the weight parameter w and starting the gradient-based optimization closer to the previously identified optimal neighborhood. Further investigation might be needed in order to refine the gradient evaluation, employ a better loss function or simply tune the gradient-based optimizers more finely.

Despite these two caveats, the gathered data seems to outline a coherent picture of the physics of the system, given the chosen trial wavefunction.

Data analysis and the interaction model

The obtained energy and α trends show how these quantities are heavily influenced by the $\tilde{\alpha}$ asymmetry parameter and the number of particles N .

In the $\tilde{\alpha} = 1.00$ case, both quantities—seen as functions of N —start from lower values than the respective non-interacting figures and are generally decreasing, except for the first few points of the energy. More precisely, the energy shows a parabolic-looking trend, while the α parameter heavily drops from $N = 2$ to $N = 4$ and even slightly increases from $N = 8$ to $N = 12$.

In the $\tilde{\alpha} = 0.40$ case, α takes on greater values than in the non-interacting case, but still proceeds to decrease as N grows. At $N = 2$, the energy is only 5% higher than the non-interacting reference and then increases less rapidly, reaching a value 8% lower than reference at $N = 8$.

Finally, in the $\tilde{\alpha} = 0.05$ case, both trends are monotonically increasing and the energy slope is greater than the non-interacting one.

Analyzing the Jastrow parameters, β_{AA} and β_{AB} , highlights how they only roughly follow the trend predicted in Subsection 2.2.2 to cancel the $\sim 1/r_{ij}^{12}$ dependence from the local energy. Indeed, evaluating Eq. 2.16 with our chosen parameters predicts:

$$\beta(\tilde{\alpha} = 1.00) \approx 1.123; \quad \beta(\tilde{\alpha} = 0.40) \approx 1.025; \quad \beta(\tilde{\alpha} = 0.05) \approx 0.833;$$

and we have found:

$$\beta_{AB}(\tilde{\alpha} = 1.00) = 1.125(3); \quad \beta_{AB}(\tilde{\alpha} = 0.40) = 1.071(2); \quad \beta_{AB}(\tilde{\alpha} = 0.05) = 0.957(6).$$

The reason behind this discrepancy is that as $\tilde{\alpha}$ reduces, the $\sim 1/r_{ij}^{12}$ potential wall becomes less steep and allows particles of different flavors to approach each other more closely. Concurrently, the well depth of the Lennard-Jones potential is attenuated, i.e., particles of different flavors are less favorably attracted towards the optimal distance² $r_{ij}^{(\min)} = 1.122$ we found in Subsection 4.1.1. Eq. 2.16 is designed to remove the $1/r_{ij}^2$ divergency from the local energy and thus β_{AB} shrinks when $\tilde{\alpha}$ does. However, a lower β_{AB} allows particles of different flavors to approach each other more closely and this clashes with the concurrent potential well reduction, which would conversely advise against particles cramming up too tightly. A variational 'tug-of-war' emerges, leading the optimizer to settle on a trade-off: the β_{AB} we ultimately measure.

This claim is further reinforced by preceding literature. In fact, McMillan [12] already demonstrated this in 1965 on 1D systems: the implemented Jastrow term $u_\beta(r_{ij}) = (\beta/r_{ij})^5$ flawlessly cures the highest-order $1/r_{ij}^{12}$ Lennard-Jones repulsive core, but it is strictly monotonically decaying. Therefore, it lacks the variational flexibility to actively accommodate the $-1/r^6$ attractive well without compromising the exact analytical divergence-canceling condition. A better suited wavefunction should comprise an enriched Jastrow factor capable of decoupling the short-range hard-core repulsion from the intermediate-range attraction. A first improvement may be sought by introducing an explicit attractive term $-(\xi/r_{ij})^\eta$. This would grant the optimizer the additional degrees of freedom needed to actively bind the particles at the equilibrium distance, completely independently of the short-range parameter β .

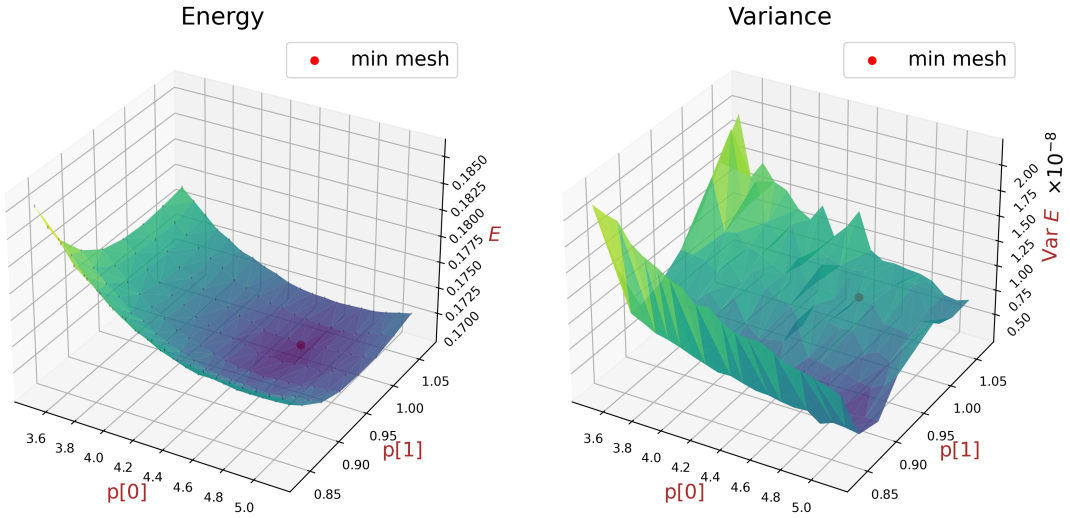


FIGURE 4.5: Mesh Grid run ($\sim 10^7$ MC steps) on an $N = 2$ system governed by a $\omega = \omega_1$ harmonic trap and a $\tilde{\alpha} = 0.05$ Lennard-Jones potential. $p[0]$ is the α parameter; $p[1]$ is the β_{AB} parameter. The minimum-energy point is highlighted in red both in the energy plot and in the variance plot. It is evident how the variance valley does not mimic the energy valley.

²Note that this value is smaller than the characteristic widths of the ground state of the harmonic confinements that we employ and therefore the Lennard-Jones potential is never—in first approximation—prevented from playing an attractive role. This heuristic should be refined, taking several other factors into account and was actually our first flaky attempt to explain the present data. For the sake of completeness it is still drafted in Appendix A.1.

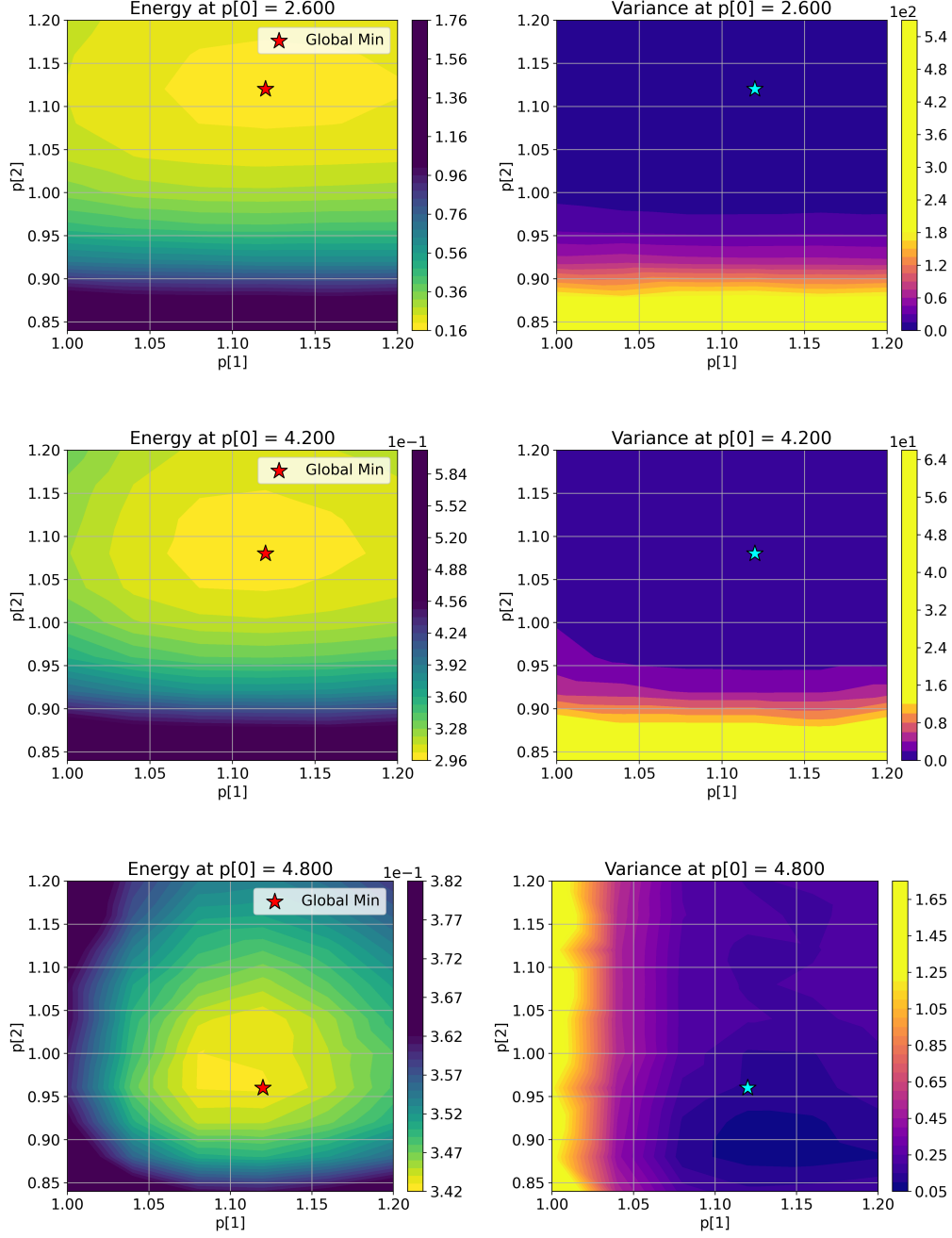


FIGURE 4.6: Comparison of Mesh Grid runs ($\sim 10^7$ MC steps) on the $N = 4$ systems governed by a $\omega = \omega_1$ harmonic trap. The top figures depict a Mesh Grid slice of the $\tilde{\alpha} = 1.00$ system; the middle figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.40$ system; the bottom figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.05$ system. $p[0]$ is the α parameter; $p[1]$ is the β_{AA} parameter; $p[2]$ is the β_{AB} parameter. The minimum-energy point is highlighted each energy plot and in the corresponding variance plot. It is evident how the variance valley does not strictly mimic the energy valley.

Energetic Balances

We can still try to interpret our data with the help of new Monte Carlo runs, outputting the partial energetic balances of each system. The resulting plots, linearly normalized by the number of particles, are shown in Figure 4.7.

All three $\tilde{\alpha}$ cases show how the Lennard-Jones attraction fights the positive contributions of the kinetic energy and of the harmonic potential. The harmonic potential is the most linear contribution and is thus impactful only in systems composed of a small number of particles. The kinetic energy shows a much more dramatic trend, especially in the $\tilde{\alpha} = 1.00$ case since the α parameter's reduction increases the momentum of the particles. In spite of this, the Lennard-Jones potential manages to prevail thanks to the combined action of both the 'alike' interactions and the 'unlike' interactions. Since the number of interactions between particles of the same flavor is

$$N_{\text{alike}} = \frac{N}{2} \left(\frac{N}{2} - 1 \right) = \frac{N^2}{4} - \frac{N}{2}$$

and the number of those between particles of different flavors is

$$N_{\text{unlike}} = \left(\frac{N}{2} \right)^2 = \frac{N^2}{4},$$

we indeed expect the total Lennard-Jones contribution to scale as $\sim N^2$ and dominate over the harmonic potential at $N \rightarrow \infty$ if the geometry and the kinetic energy of the systems allow it.

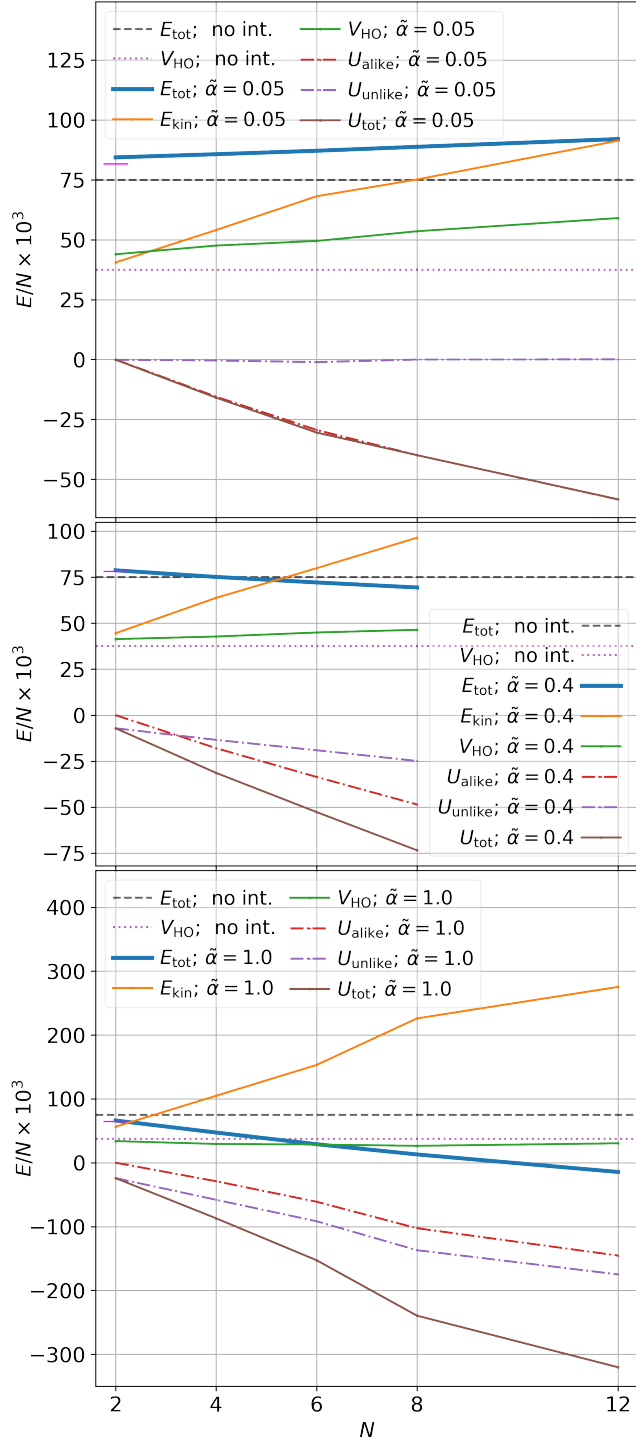


FIGURE 4.7: Partial energetic balances per particle of the strongly-confined interacting systems vs. the number of particles N . The two-body benchmark reference is also marked in magenta. FinalMC ran with $2^{24} \sim 10^7$ MC cycles.

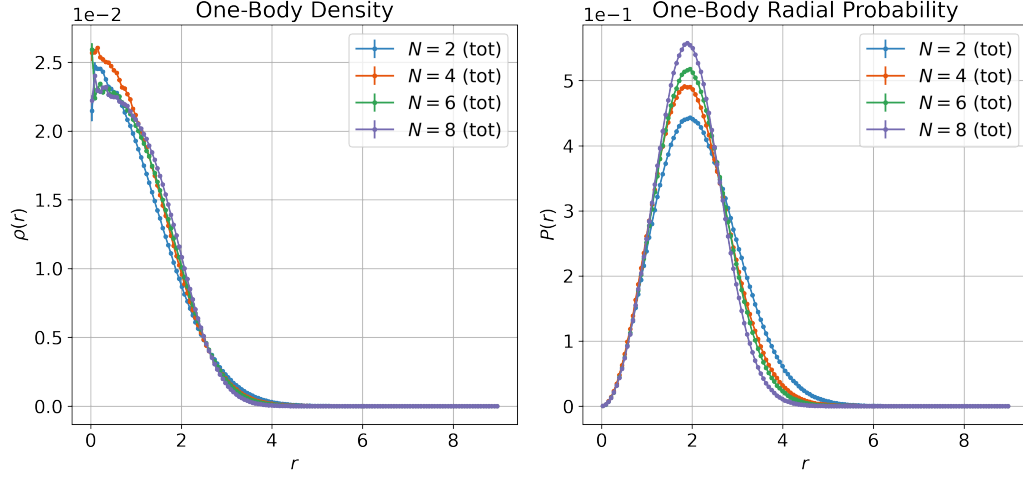


FIGURE 4.8: Radial one-body densities ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. The plots are normalized by the number of particles N .

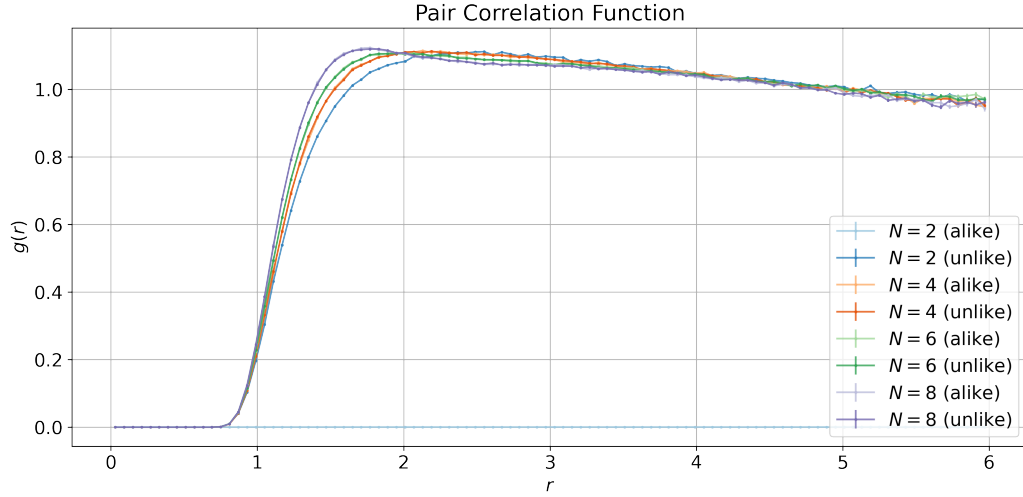


FIGURE 4.9: Radial pair correlation function ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential.

Looking at the one-body density plots in Figure 4.8, one can notice how the increased number of particles visually tightens the overall distribution in the $\tilde{\alpha} = 1.00$ case. The PCFs in Figure 4.9 show how the slight increase in the β parameters, observed in the same $\tilde{\alpha} = 1.00$ systems when N grows from 2 to 8, results in progressively correlated particles—regardless of their relative flavors—even when it should be unfavorable for the divergence-canceling condition. Keep in mind that this effect might appear more definite in this representation than it really is, due to the errors affecting the β parameters. More details about the spatial distributions of $\tilde{\alpha} = 1.00$ systems follow the comparisons referencing Figures 4.12 and 4.13.

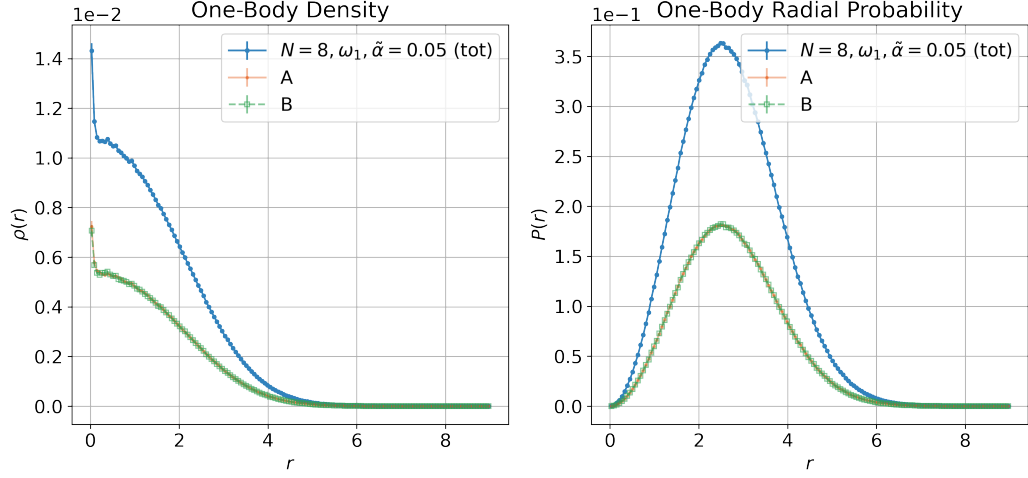


FIGURE 4.10: Radial one-body density ($\sim 10^7$ MC steps) of the $N = 8$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.05$ Lennard-Jones potential. OBDs of the A particles and B particles are also individually plotted. All plots are normalized by the same number of particles $N = 8$. It is evident that the OBD of species A perfectly overlaps that of species B , and each contributes exactly half of the total $\rho(r)$ as expected (Subsection 2.4.1).

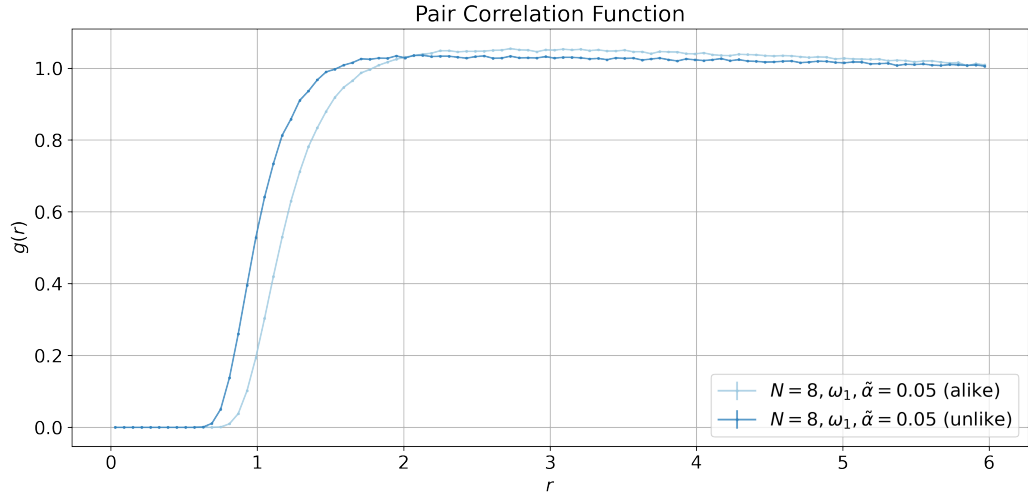


FIGURE 4.11: Radial pair correlation functions ($\sim 10^7$ MC steps) of the $N = 8$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.05$ Lennard-Jones potential.

Analyzing the PCFs in Figure 4.11, we see that the $\beta_{AB} < \beta_{AA}$ parameters found in the $\tilde{\alpha} = 0.05$ systems draw particles of different flavors together to cure the less steep $1/r_{ij}^{12}$ divergence, instead of distancing them according to the shallower potential well.

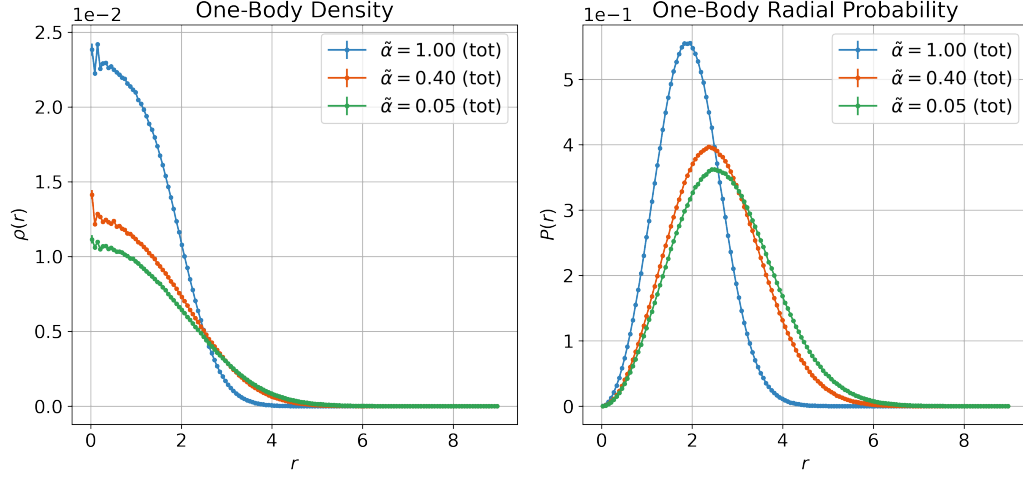


FIGURE 4.12: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap. The plots are normalized by the number of particles. It is evident how a stronger Lennard-Jones interaction between different species allows for an overall more compact system.

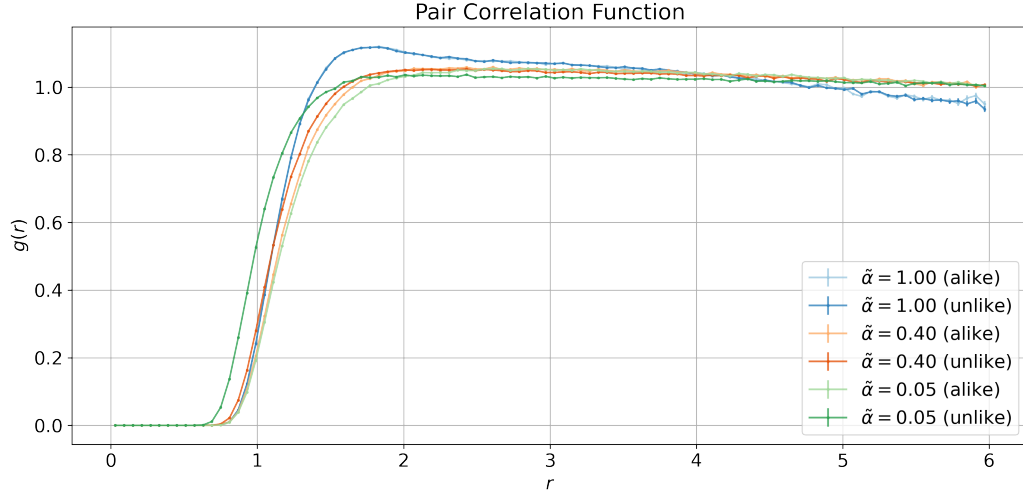


FIGURE 4.13: Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap.

Finally, Figures 4.12 and 4.13 further demonstrate that the $\tilde{\alpha} = 1.00$ system is significantly more cohesive than the $\tilde{\alpha} < 1$ systems and provide the first hint of a transition to the liquid state (i.e., $g(r)$ distinctively peaks above $g = 1$ before stabilizing at the $g(\infty) = 1$ asymptote).

4.3.2 Weak confinement

Now we proceed by running the VMC code setting $\omega = \omega_2 = 0.027951$.

The results shown in Table 4.5 and Figure 4.14 reinforce everything we have stated so far even more firmly.

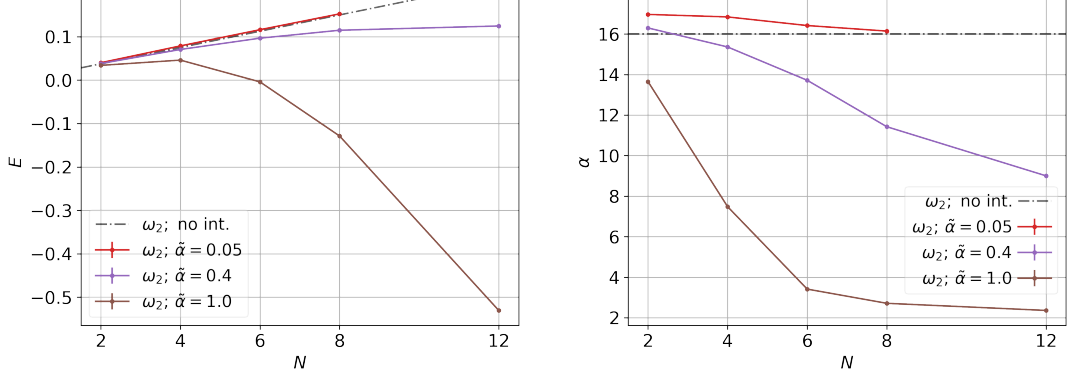


FIGURE 4.14: Best energy and α values obtained for the interacting systems inside a $\omega = \omega_2 = 0.027951$ harmonic confinement.

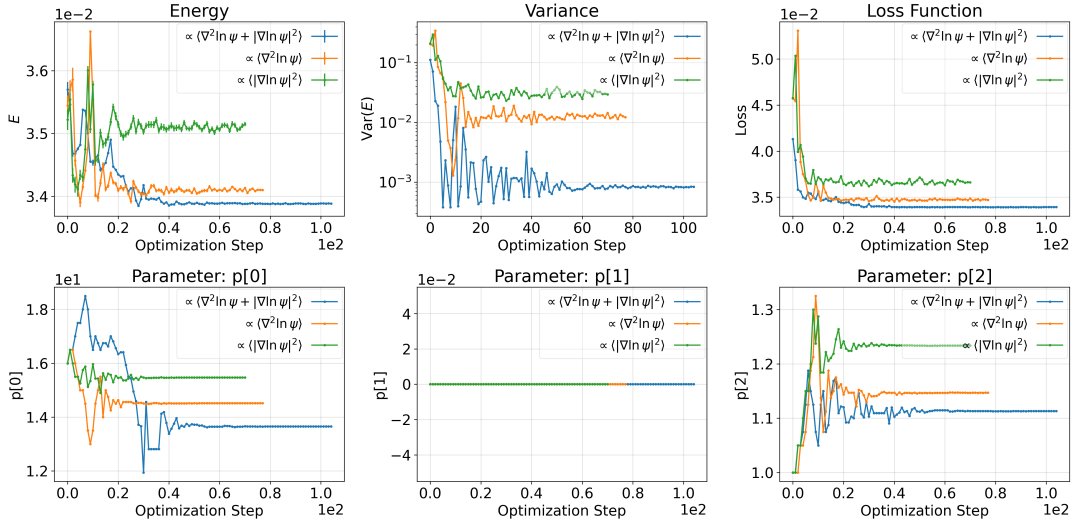


FIGURE 4.15: Comparison of Nelder-Mead optimization processes for the $N = 2$ interacting system inside a $\omega = \omega_2$ harmonic confinement. The blue optimization utilized Eq. 2.19 to evaluate the local kinetic energy; the orange one utilized Eq. 2.20; the green one utilized Eq. 2.21. Parameters $p[0]$, $p[1]$ and $p[2]$ correspond to α , β_{AA} and β_{AB} , respectively.

TABLE 4.5: VMC results for interacting systems inside a $\omega = \omega_2$ harmonic confinement. Different optimization methods were tested ("N-M" stands for "Nelder-Mead"; "M-G" stands for "Mesh Grid"). The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.50, 0.05, 0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 8×10^6 MC cycles. Adam ran 1000 steps with 5×10^5 MC cycles, an initial learning rate of 0.05 and `autoMCincr.wait` = 500. Both NLOPT optimization algorithms used `x_tol` = 10^{-5} as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values.

N	$\tilde{\alpha}$	$E^{(\text{exp.})}$	Optim.	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
2	1.00	0.030 91	N-M	0.05	16	—	1	13.651	—	1.1130	0.033 880(4)
2	1.00	0.030 91	N-M [†]	0.05	16	—	1	14.513	—	1.1469	0.034 10(1)
2	1.00	0.030 91	N-M [‡]	0.05	16	—	1	15.471	—	1.2336	0.035 13(2)
2	1.00	0.030 91	M-G	0.00	—	—	—	13.7(2)	—	1.12(6)	0.033 89(1)
4	1.00	—	N-M	0.05	16	1.1	1.1	7.4787	1.1127	1.1165	0.046 00(2)
4	1.00	—	M-G	0.00	—	—	—	7.5(5)	1.10(5)	1.10(5)	0.0458(1)
6	1.00	—	N-M	0.05	16	1.1	1.1	3.5038	1.1241	1.1228	−0.004 02(2)
6	1.00	—	N-M**	0.05	3.5038	1.1241	1.1228	3.4118	1.1236	1.1207	−0.004 16(6)
6	1.00	—	M-G	0.00	—	—	—	4(1)	1.1(1)	1.1(1)	−0.0022(4)
8	1.00	—	N-M	0.05	16	1	1	2.7701	1.1296	1.1268	−0.1282(1)
8	1.00	—	N-M**	0.05	2.7701	1.1296	1.1268	2.7110	1.1239	1.1261	−0.1287(1)
8	1.00	—	M-G	0.00	—	—	—	3(1)	1.1(1)	1.1(1)	−0.1211(9)
12*	1.00	—	N-M	0.05	2	1.1	1.1	2.3599	1.1301	1.1314	−0.5304(2)
12*	1.00	—	M-G	0.00	—	—	—	2.5(5)	1.12(4)	1.12(4)	−0.5269(10)
2	0.40	0.037 92	N-M	0.05	16	—	1	16.298	—	1.0666	0.038 021(2)
2	0.40	0.037 92	M-G	0.00	—	—	—	16.5(5)	—	1.06(6)	0.038 01(1)
4	0.40	—	N-M	0.05	16	1.1	1.1	15.360	1.1156	1.0697	0.070 471(6)
4	0.40	—	M-G	0.00	—	—	—	15.0(5)	1.12(6)	1.06(6)	0.070 48(4)
6	0.40	—	N-M	0.05	13	1.1	1	13.716	1.1108	1.0646	0.096 46(2)
6	0.40	—	M-G	0.00	—	—	—	13.0(5)	1.12(6)	1.06(6)	0.096 48(8)
8	0.40	—	N-M	0.05	16	1	1	11.424	1.1106	1.0706	0.114 76(4)
8	0.40	—	M-G	0.00	—	—	—	12.0(5)	1.12(6)	1.06(6)	0.1154(1)
12*	0.40	—	Adam	0.0005	2	1.1	1.1	9.0597	1.1011	1.0814	0.1248(1)
12*	0.40	—	BFGS	0.0005	2	1.1	1.1	9.0017	1.1025	1.0835	0.1246(1)
12*	0.40	—	M-G	0.00	—	—	—	9.0(5)	1.08(4)	1.06(6)	0.126(2)
2	0.05	0.039 06	N-M	0.05	17	—	1	16.966	—	0.9459	0.039 984(7)
2	0.05	0.039 06	M-G	0.00	—	—	—	17.5(5)	—	1.00(5)	0.0399(4)
4	0.05	—	N-M	0.05	16	1	1	16.842	1.1051	0.9411	0.078 69(2)
4	0.05	—	M-G	0.00	—	—	—	17.0(5)	1.12(6)	1.00(5)	0.0784(2)
6	0.05	—	N-M	0.05	16	1.1	1	16.417	1.1083	0.9875	0.116 01(5)
6	0.05	—	M-G	0.00	—	—	—	16.5(5)	1.12(6)	0.95(10)	0.1164(2)
8	0.05	—	N-M	0.05	16	1.1	1	16.143	1.1192	0.9559	0.152 57(8)
8	0.05	—	M-G	0.00	—	—	—	16.5(5)	1.12(6)	0.95(10)	0.1525(3)

[†] Local kinetic energy evaluated with Eq. 2.20.

[‡] Local kinetic energy evaluated with Eq. 2.21.

* Bonus run.

** These runs were executed subsequently to those of Table 4.6 for the sake of fair comparison.

Because the harmonic trap is substantially broader ($\alpha = 16$), the harmonic energy barrier is significantly lowered. Consequently, all observed regimes display a globally much more cohesive behavior: across all datasets, as N grows, α decreases and so does the total energy per particle E/N plotted in Figure 4.16.

Once again, the two-body systems exhibit somewhat higher energy estimates compared to the respective predictions of Table 4.2 (+9.6% in the $\tilde{\alpha} = 1.00$ case, +0.3% in the $\tilde{\alpha} = 0.40$ case and +2.4% in the $\tilde{\alpha} = 0.05$ case).

Likewise, the Jastrow parameters again reveal the variational ‘tug-of-war’. The weak cross-interactions ($\tilde{\alpha} = 0.05$ and $\tilde{\alpha} = 0.40$) still drive the optimizer to slightly inflate β_{AB} above the theoretical predictions in an effort to push the uncoupled species apart while concurrently satisfying the divergence-canceling condition.

If a more flexible wavefunction had been employed, here one might have possibly observed a species separation in the $\tilde{\alpha} = 0.40$ and $\tilde{\alpha} = 0.05$ cases: while the β_{AB} parameter would have taken care of the less steep $\sim 1/r_{ij}^{12}$ repulsion wall exclusively, the additional attractive Jastrow term could have kept particles of different flavors apart at longer distances in accord with the shallower well.

Finally, this regime, together with the previous ($N = 12$, $\omega = \omega_1$, $\tilde{\alpha} = 1.00$) system, unveils a striking result: for $\tilde{\alpha} = 1.00$, the system undergoes a crossover from a trapped quantum gas into a self-bound quantum liquid droplet.

As the particle number N increases, the α parameter collapses dramatically from $\alpha \approx 13.6$ down

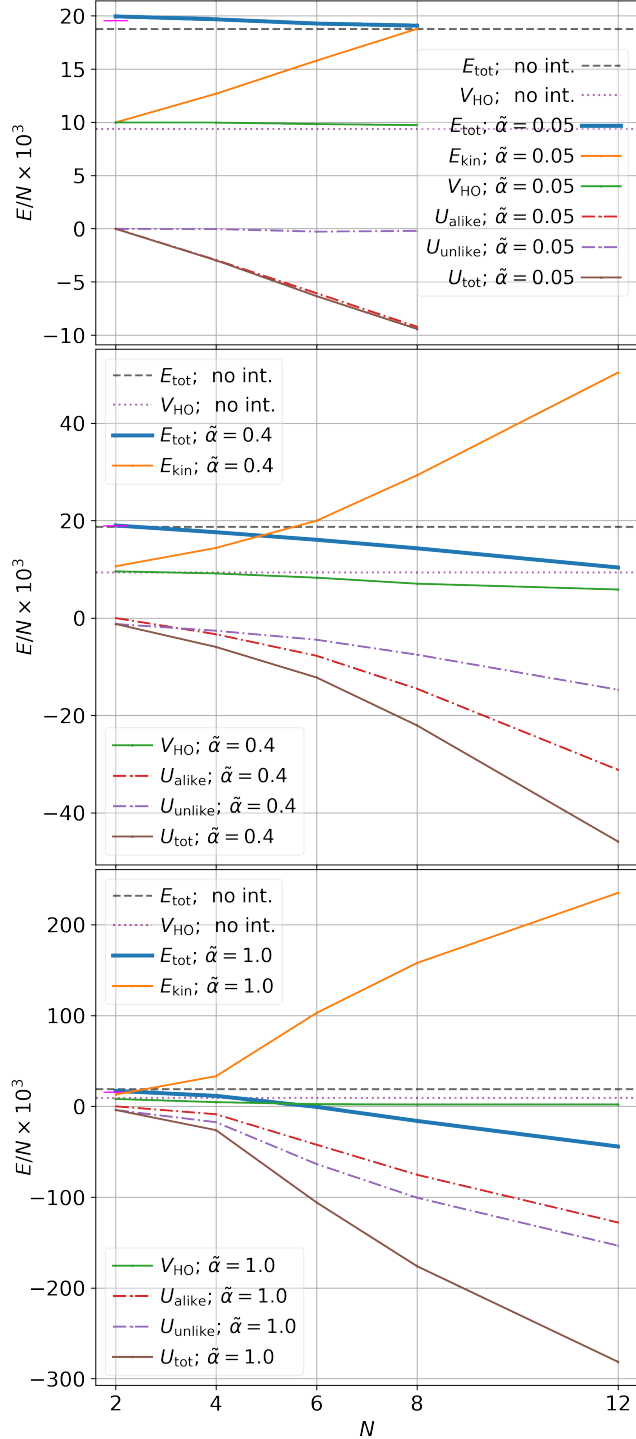


FIGURE 4.16: Partial energetic balances per particle of the weakly-confined interacting systems vs. the number of particles N . The two-body benchmark reference is also marked in magenta. FinalMC ran with $2^{24} \sim 10^7$ MC cycles.

to $\alpha \approx 2.7$ and, for $N \geq 6$, the total ground-state energy crosses below zero. In Figure 4.16 we can also see how the Gaussian width collapse drastically lowers the harmonic potential per particle from the constant trend of the non-interacting case. This simultaneously raises the momentum considerably—and thus also the kinetic energy—to compensate for the increased locality of the system. The same phenomenon was already present in the strongly-confined system, but here it is much more evident.

Furthermore, since both the expected values of the quantum kinetic energy and the external harmonic potential are strictly positive quantities, measuring an overall negative energy indicates that the cohesive Lennard-Jones attraction already dominates the trapped system's energetics; we show in the next subsection that this in fact rigorously implies self-binding even if the confinement is removed ($\omega = 0$).

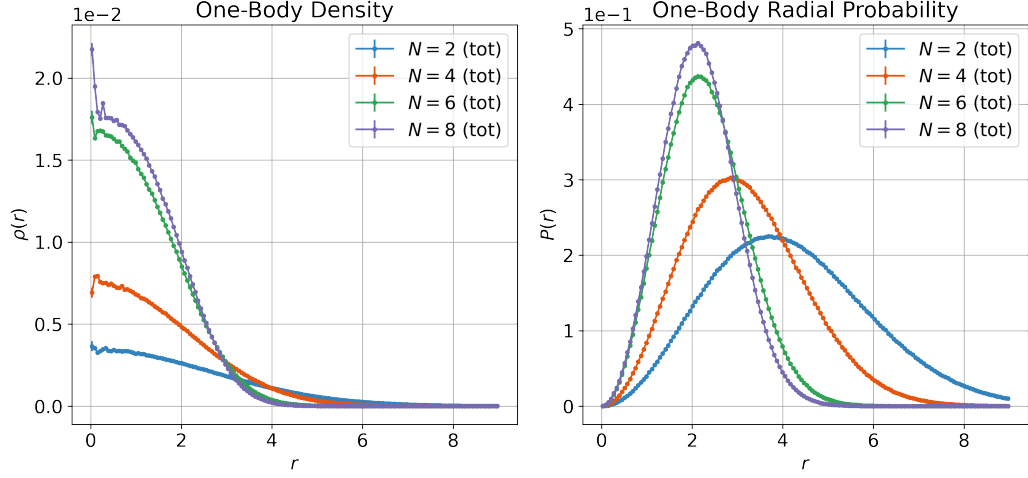


FIGURE 4.17: Radial one-body densities ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_2$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. The plots are normalized by the number of particles.

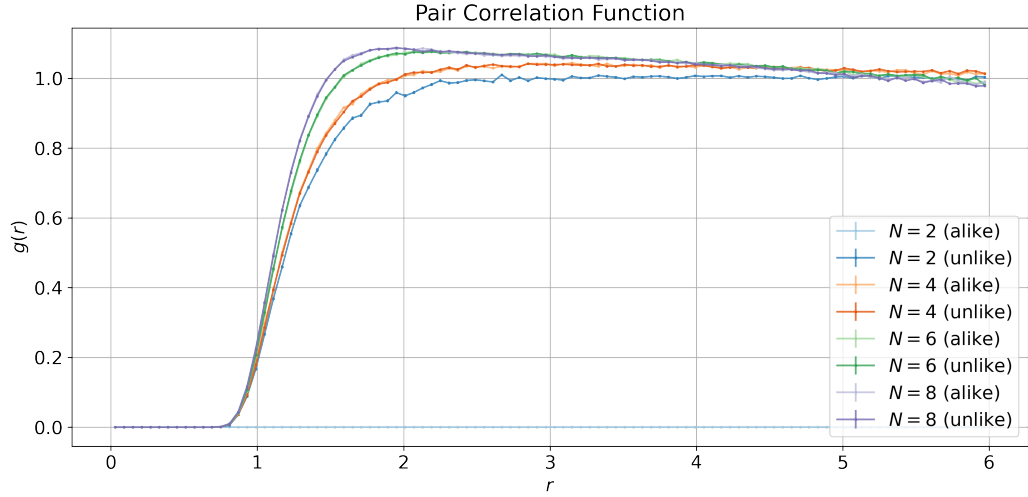


FIGURE 4.18: Radial pair correlation function ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_2$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential.

The one-body densities plotted in Figure 4.17 clearly visualize the Gaussian width collapse. Moreover, Figure 4.18 outlines a gas-like behavior in the PCF of the $N = 2$ (i.e., $g(r)$ monotonically rises to the $g(\infty) = 1$ asymptote) system and a liquid-like behavior in the PCF of the $N = 8$ system (i.e., $g(r)$ distinctively peaks above $g = 1$ before stabilizing at the $g(\infty) = 1$ asymptote), although multiple oscillations around the asymptote cannot be observed employing this wavefunction (cf. [11]). Even employing a better wavefunction, one might possibly struggle to observe oscillations due to the small size of the clusters and the presence of the harmonic confinement.

4.3.3 No confinement

We can benchmark this structural stability further by running the VMC program one last time: we set $\omega = 0$ —effectively removing the harmonic confinement—and initiate the optimization process from the optimal parameters found for the $\omega = \omega_2, \tilde{\alpha} = 1.00$ systems.

While the $N \geq 6$ systems remained completely intact, the $N = 2$ and $N = 4$ systems struggled to find a minimum and resorted to disaggregating the bosonic cloud as much as possible³. The results for the systems that successfully converged are shown in Table 4.6.

N	$\tilde{\alpha}$	Optim.	w	$\alpha^{(0)}$	$\beta_{AA}^{(0)}$	$\beta_{AB}^{(0)}$	α	β_{AA}	β_{AB}	E
6	1.00	Nelder-Mead	0.05	3.5038	1.1241	1.1228	3.7838	1.1220	1.1184	-0.019 28(6)
6	1.00	M-G	0.00	—	—	—	3.7(7)	1.12(4)	1.12(4)	-0.0189(2)
8	1.00	Nelder-Mead	0.05	2.7701	1.1296	1.1268	2.8182	1.1249	1.1260	-0.1459(1)
8	1.00	Mesh Grid	0.00	—	—	—	3.0(3)	1.12(4)	1.12(4)	-0.1465(4)
12*	1.00	Nelder-Mead	0.05	2.3	1.12	1.12	2.5078	1.1336	1.1298	-0.5544(2)
12*	1.00	Nelder-Mead	0.05	2.3	1.12	1.12	2.4242	1.1291	1.1286	-0.5561(2)
12*	1.00	Mesh Grid	0.00	—	—	—	2.3(3)	1.12(4)	1.12(4)	-0.551(1)

* Bonus run.

TABLE 4.6: Successful VMC results for interacting systems without any confinement. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm[0.50, 0.05, 0.05]$ as initial state setting, $\times 10^7$ MC cycles and used $\text{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values.

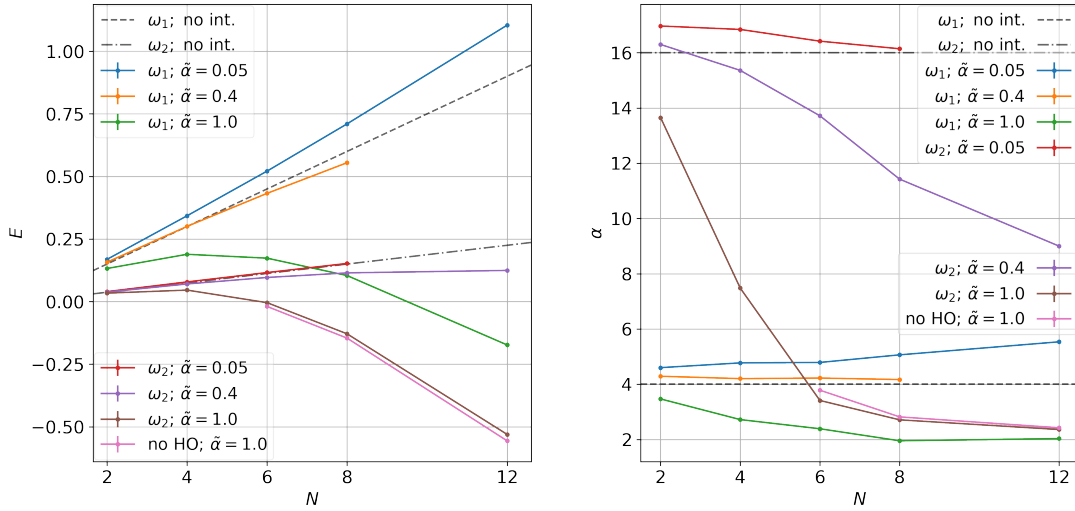
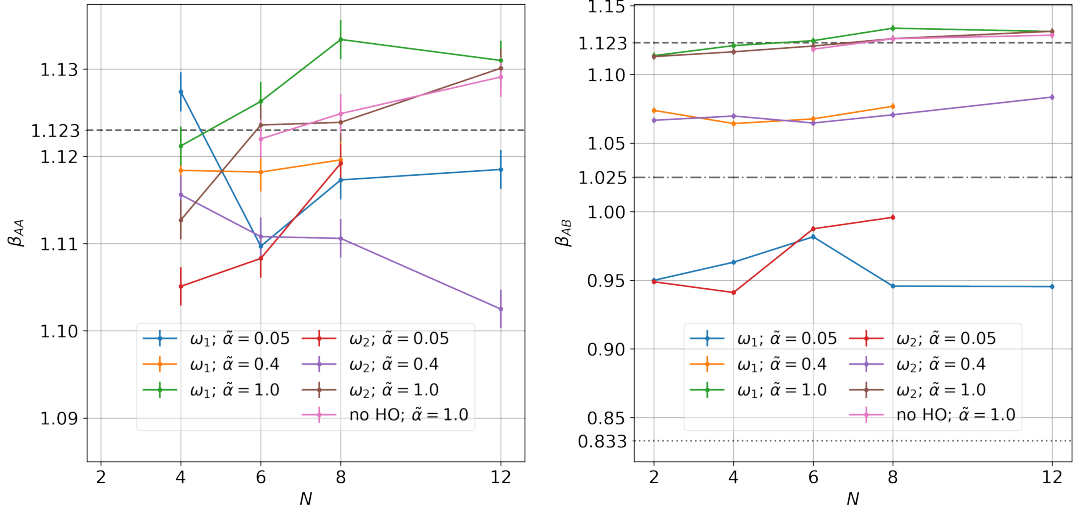


FIGURE 4.19: Summary of the energy and α values obtained for the interacting systems.

³Virtually $\alpha \rightarrow \infty$, though the numerical upper bound was set to $\alpha = 50$. Furthermore, we ran the same tests for the ($N = 8, 12$; $\tilde{\alpha} = 0.40$) systems and, likewise, they did not converge to any finite α .

FIGURE 4.20: Summary of the β_{AA} and β_{AB} values obtained for the interacting systems.

The two failure modes reported above are physically distinct.

For $N = 2$, we can check directly: when we solve the free-space two-body relative-motion equation of Subsection 4.1.3 without the harmonic term, the lowest eigenvalue does not seem to converge to a stable negative value as the numerical domain is enlarged, but instead shrinks steadily toward zero—the signature of a state with no true bound state, rather than a converging bound-state energy. This is consistent with the system’s de Boer parameter, $\Lambda^2 = \hbar^2/(m\sigma^2\epsilon) = 0.2$, being slightly larger (i.e., more quantum) than that of real ^4He ($\Lambda^2 \approx 0.18$), whose dimer is itself only marginally bound (approx. 1 mK, [5]). The divergence of α for $N = 2$ is therefore the expected numerical fingerprint of a genuinely unbound system at this mass, not an optimizer failure.

The $N = 4$ case is different: a free ^4He tetramer is known to be comfortably bound (approx. 0.55 K, [6, 10]). The non-convergence here more likely reflects—once again—a limitation of the trial wavefunction. A very weakly-bound cluster’s true wavefunction develops a long, slowly-decaying tail set by its (small) binding energy, whereas the Gaussian one-body factor $e^{-r_i^2/2\alpha}$ decays much faster than any such tail regardless of α , and is structurally unable to represent it—the same kind of limitation already identified for the Jastrow factor in Subsection 4.3.1, here showing up in the radial part of the ansatz instead.

For $N = 6, 8, 12$, as hinted above, self-binding can in fact be established without the free-space optimization at all. Since the non-confined Hamiltonian is such that $H(0) = H(\omega) - V_{\text{HO}}(\omega)$ and $V_{\text{HO}}(\omega) = \frac{1}{2}\omega^2 R^2 \geq 0$ for any configuration $\mathbf{R}$, the already-optimized trapped wavefunction $\psi_T^*(\omega)$ is itself a valid (if suboptimal) trial state for $H(0)$. The variational principle then gives

$$E_{\text{free}} \leq \langle H(0) \rangle_{\psi_T^*(\omega)} = E_{\text{VMC}}(\omega) - \langle V_{\text{HO}} \rangle_{\psi_T^*(\omega)} < E_{\text{VMC}}(\omega),$$

where E_{free} is the true ground-state energy of the non-confined system, and $E_{\text{VMC}}(\omega)$ are the energies computed via VMC for the previous trapped systems. This implies that any ω for which $E_{\text{VMC}}(\omega) < 0$ already guarantees $E_{\text{free}} < 0$.

Table 4.5 shows exactly this for $N = 6, 8, 12$ at ω_2 and so does Table 4.4 for $N = 12$ at ω_1 , hence self-binding for these systems is established rigorously, even independently of the direct $\omega = 0$ optimization; the results in Table 4.6 serve as an independent confirmation and additionally supply the (tighter) actual free-space energy values.

A complementary, fully self-contained check is *stability against fragmentation*: a cluster is stable against splitting into two sub-clusters of size k and $N - k$ when

$$E(N) < E(N - k) + E(k).$$

Using only the $\omega = 0$ values in Table 4.6, $E(12) = -0.5544 \ll E(6) + E(6) = -0.0386$, confirming that the $N = 12$ droplet is stable against splitting into two $N = 6$ droplets.

In this limit, the droplet's spatial distribution is governed entirely by the internal Lennard-Jones balance rather than the external trap walls: the $N = 6, 8, 12$ clusters form a cohesive, self-bound liquid that would remain structurally stable even in the limit of vanishing confinement.

Finally, comparing the α and energy trends in Figure 4.19, we see that the α curves seem to head towards a common asymptote for $N \rightarrow \infty$, signifying the incompressibility of the liquid; whereas the energies $E(\omega; N)$ appear to drop while maintaining a constant difference between the per-particle energy estimates $\varepsilon(\omega) = \frac{E(\omega; N) - E(0; N)}{N}$:

$$\varepsilon(\omega_1) = 0.0318(2) \quad \text{and} \quad \varepsilon(\omega_2) = 0.0023(1).$$

This constancy is not merely coincidental. Since $\langle H(0, N) \rangle_{\psi(\alpha)} = \langle H(\omega, N) \rangle_{\psi(\alpha)} - \langle V_{\text{HO}} \rangle_{\psi(\alpha)}$ for a fixed trial wavefunction $\psi(\alpha)$, and the optimal trapped and free-space wavefunctions are found to be nearly identical for $N \geq 6$ (Table 4.6 and Figures 4.21 and 4.22), $\varepsilon(\omega) \approx \langle V_{\text{HO}} \rangle / N$ directly. Using the corresponding exact $\langle V_{\text{trap}} \rangle$ estimates obtained from the energetic-balance runs and displayed in Figures 4.7 and 4.16, one obtains the following mean values:

$$\text{mean}(\langle V_{\text{HO}} \rangle / N)(\omega_1) = 0.028(2) \quad \text{and} \quad \text{mean}(\langle V_{\text{HO}} \rangle / N)(\omega_2) = 0.0022(1).$$

These roughly match the $\varepsilon(\omega_1)$ and $\varepsilon(\omega_2)$ values reported above, especially at ω_2 . Ultimately, this confirms that $\varepsilon(\omega)$ is, to good approximation, simply the harmonic-trap energy per particle evaluated on the (nearly N -independent, for the largest N considered) self-bound wavefunction, rather than a separately-emerging constant—consistent with an incompressible liquid whose internal structure no longer depends on N in this regime.

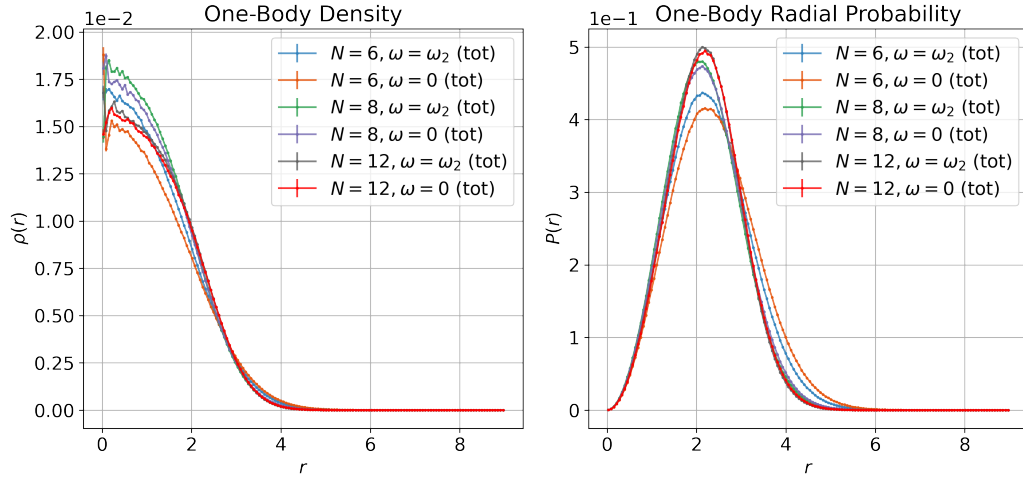


FIGURE 4.21: Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. The plots are normalized by the number of particles. All the quantum droplets exhibit an approximately equal internal structure.

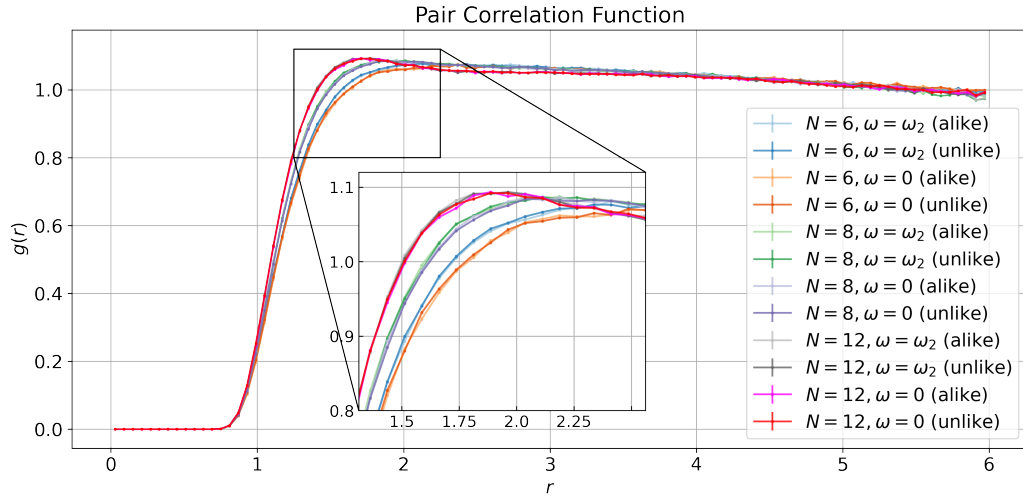


FIGURE 4.22: Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. All the quantum droplets exhibit an approximately equal internal structure.

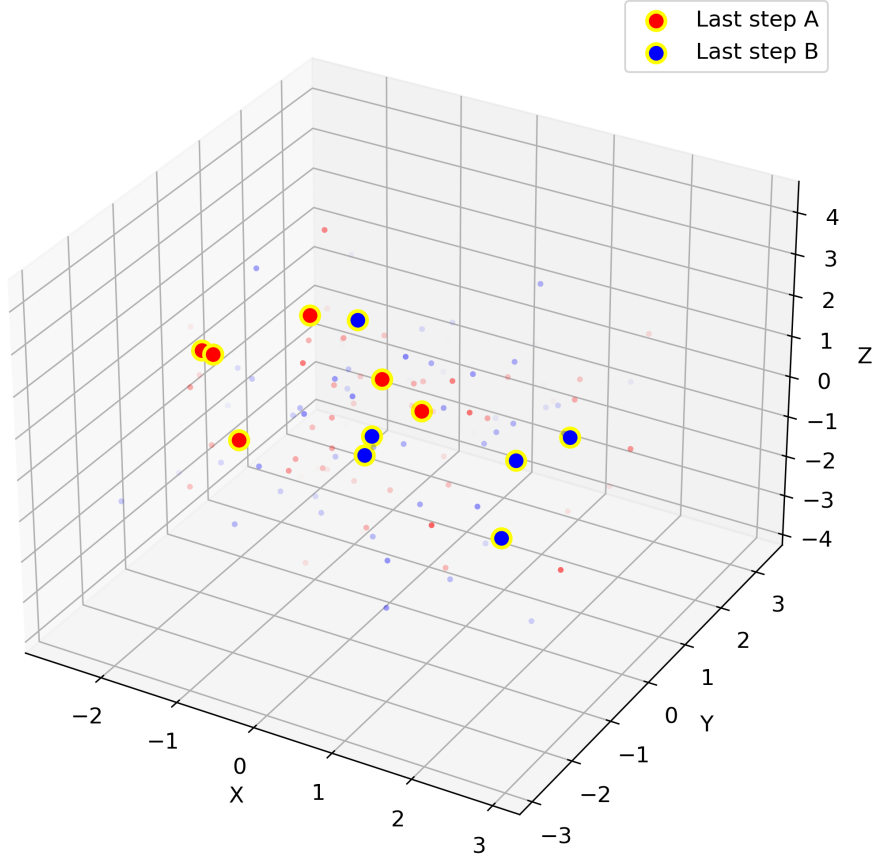


FIGURE 4.23: 3D visualization of the particles' positions over the course of 10 non-subsequent Metropolis steps (evenly extracted from a total of 2×10^7 steps) for the $N = 12$ non-confined system. The last configurations is highlighted, while the 9 preceding ones are progressively faded.

Conclusions

The Variational Monte Carlo method has proven to be a valuable computational framework for investigating the complex few-body physics ($N = 2, 4, 6, 8, 12$) of the systems analyzed in this thesis. Through its robust implementation, we sought to characterize the ground states (0 K) of a binary bosonic mixture affected by a Lennard-Jones potential. The case of symmetric interaction ($\tilde{\alpha} = 1.00$) corresponds to a cluster of ${}^4\text{He}$ atoms.

This study faced the same problems encountered by McMillan [12] (here in 3D): the chosen trial wavefunction has not been fully capable of representing the desired true ground states. This limitation could only be quantitatively benchmarked for $N = 2$, where an essentially exact two-body reference is available; for $N \geq 4$, no external benchmark was attempted, and a direct comparison against literature estimates for small ${}^4\text{He}_N$ clusters—available at least for the symmetric-interaction case, $\tilde{\alpha} = 1.00$ —would be needed to establish how the wavefunction’s bias scales with the number of particles. Nevertheless, the coded VMC program—which makes extensive use of well-known computational techniques and couples them with automated routines—was successfully benchmarked and upper bounds to the ground states were determined. These were used to gain insight into the physics of the system. Most notably, we simulated the interaction-driven crossover from a confined trapped gas to a self-bound quantum liquid droplet, governed purely by the delicate balance of the Lennard-Jones potential.

This work highlights several directions for future improvements.

Firstly, the Metropolis auto-tuning of the `step` could be upgraded. The relative frequency at which a flavor-swap move is proposed instead of a standard spatial step may be adjusted as a second degree of freedom on which to act in order to achieve the desired acceptance ratio and ensure an overall better ergodic sampling. To do so, one should first measure the acceptance ratio of each respective kind of proposal individually.

Secondly, while the employed optimization algorithms represented a solid alternative to brute-force manual searches—speeding up the data collection process and laying a common standard for result comparison—they are far from fail-proof and leave ample margin for improvement. The program and the physical outcomes would greatly benefit from advanced, noise-resilient optimization algorithms (such as Stochastic Reconfiguration) specifically tailored for quantum Monte Carlo energy landscapes.

On a similar note, as previously mentioned, the physics of this system could have been better expressed by a finer wavefunction choice. In particular, a purely pairwise correlation factor of the kind used here is structurally limited in the accuracy it can reach: subsequent extensions typically introduce explicit three-body (triplet) correlation terms, parametrized over three-particle cluster coordinates, to capture correlations that no two-body Jastrow factor can represent regardless of how its parameters are tuned. The most recent state-of-the-art VMC codes exploit Neural Quantum States in place of fixed Jastrow-based wavefunctions,

CONCLUSIONS

as discussed in Project [2].

Finally, the same physical system could be investigated from entirely different perspectives. For instance, one could improve the ground-state accuracy by employing Diffusion Monte Carlo (DMC) techniques, which project out the exact ground-state via imaginary-time evolution. Alternatively, one could simulate the real-time dynamics of the droplets following a sudden trap release to study non-equilibrium fragmentation [3], or alter the system radically by replacing bosons with fermions, thus comparing the fundamentally different behaviors dictated by the Pauli exclusion principle in fully anti-symmetric systems.

Appendix A

Appendix

A.1 Heuristical Analysis of the Confined Interacting Systems

A first explanation for the data observed in the Strongly Interacting System can be sought by analyzing the two-body problem and subsequently extending the reasoning to a larger number of particles.

From Eq. 4.1, we know the minimum of the Lennard-Jones interaction is achieved once the pair of particles is found at a relative distance of $r_{ij} \approx 1.122$. This value is smaller than the Gaussian width $\sqrt{\alpha}$ imposed on all the particles by the harmonic confinement. Heuristically speaking, we can thus claim that the Lennard-Jones potential tends to draw particles closer together than the natural width of the trap, which favors a smaller spatial variance ($\alpha < 4$). In practice, we observe that the results obtained for the $N = 2$ systems show that this holds true only for $\tilde{\alpha} = 1.00$, whereas the $\tilde{\alpha} = 0.40$ and $\tilde{\alpha} = 0.05$ systems exhibit the very opposite behavior. This is because we need to take into account the Zero Point Energy of the 2-particle non-interacting system and compare it to the depth of the Lennard-Jones potential well we found in Eq. 4.2.

From the two-body Hamiltonian of Eq. 4.3, we know that the ground state energy of the center of mass system is $E_{CM} = \frac{3}{2}\hbar\omega = E_0$ and that the relative motion of the two particles in 3D has a ZPE of $E_{1,2} = \frac{3}{2}\hbar\omega = E_0$. Given our current choice $\omega = \omega_1$, this means that $\tilde{\alpha} \gtrsim E_0 \approx 0.075$ must hold for a pair of particles of different flavors to be bound together by the Lennard-Jones interaction; otherwise, they are repelled by the $\sim 1/r_{ij}^{12}$ potential wall. Likewise, all pairs of particles alike in flavor show an attractive behavior since $\epsilon = 1 \gg E_0 = 0.075$.

The next step, as anticipated earlier, is to address how the overall effect of the Lennard-Jones potential on the system varies as the number of particles grows. The number of interactions between particles of the same flavor is

$$N_{\text{alike}} = \frac{N}{2} \left(\frac{N}{2} - 1 \right) = \frac{N^2}{4} - \frac{N}{2}$$

and the number of those between particles of different flavors is

$$N_{\text{unlike}} = \left(\frac{N}{2} \right)^2 = \frac{N^2}{4}.$$

Additionally we know that the harmonic contribution to the energy grows linearly with N .

Now we may—only approximately—explain the $E(N)$ and $\alpha(N)$ trends observed for the $\tilde{\alpha} = 1.00$ and $\tilde{\alpha} = 0.05$ cases, and partly for the $\tilde{\alpha} = 0.40$ case.

- In the $\tilde{\alpha} = 1.00$ case, the $N = 2$ system is composed of only one (A, B) attractive pair (because $\tilde{\alpha} > E_0$) and no 'alike' pairs. $\alpha \approx 3.5 < 4$ is found, hence the particles are more tightly bound than in the respective non-interacting system. Furthermore, as the number of particles N grows, the system is more and more tightly bound since the number of attractive interactions scales $\propto (N_{\text{unlike}} + N_{\text{alike}}) \sim \frac{N^2}{2}$, unlike the $\propto N$ harmonic confinement contribution. In fact, α suddenly drops, but subsequently plateaus. Conversely, the energy slightly increases until $N = 6$ and then starts falling (almost outlining the predicted parabolic trend, but there would not be any valid first-hand reason for the decrease should not start right from $N = 2$).
- In the $\tilde{\alpha} = 0.05$ case, the $N = 2$ system is composed only of one (A, B) repulsive pair (because $\tilde{\alpha} < E_0$) and no 'alike' pairs. $\alpha \approx 4.5 > 4$ is found, hence the particles are more dispersed than in the respective non-interacting system. Furthermore, as the number of particles N grows, the system seems to keep slowly enlarging. Naively assuming that each repulsive interaction exactly cancels one attractive interaction would model a $\propto (N_{\text{unlike}} - N_{\text{alike}}) = \frac{N}{2}$ linear growth, as opposed to the $\propto (N_{\text{unlike}} + N_{\text{alike}}) \sim \frac{N^2}{2}$ quadratic shrinking of the totally attractive $\tilde{\alpha} = 1.00$ case. In this case, the observed data may even be superficially explained this way. However, a more refined analysis, like the one conducted analyzing the true partial energetic balances in Subsection 4.3.1, making reference to the plots in Figure 4.7, shows how very initial claim of the existence of repulsive interactions is disproven, even before addressing the flaw in the trial wavefunction.
- The $\tilde{\alpha} = 0.40$ dataset is even harder to motivate: although the N_{unlike} pairs should contribute in a lightly attractive manner ($\tilde{\alpha} > E_0$) and α undeniably shrinks as N grows, all analyzed systems show $\alpha > 4$. Nevertheless, the $\alpha(N)$ trend is descending, remarking the somewhat attractive behavior of all the interaction pairs.

All things considered, this heuristic interaction model fails to explain too many important aspects of the observed data and must thus be discarded.

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List of Figures

4.1	Estimated standard error vs number of blocking transformations via the automated blocking algorithm (Subsection 3.2.4). The first plot analyzes the final MC (2^{27} MC cycles) run for the $N = 6$, ω_1 non-interacting system (Table 4.3); the second plot refers to a final MC run for the same system but manually setting a non-optimal $\alpha = 4.5$, resulting in an energy estimate of 0.45311 ± 0.00001 as opposed to the naive pre-blocking result 0.453107 ± 0.000002	31
4.2	Radial one-body densities ($\sim 10^7$ MC steps) of the non-interacting systems. As expected, these non-normalized plots scale with the number of particles and perfectly match the Gaussian widths $\sqrt{\alpha} = 2, 4$	32
4.3	Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 4$ non-interacting systems. The plot is normalized by the number of particles. The stronger the confinement ω , the tighter the distribution.	33
4.4	Best energy, α and β values obtained for the interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement. The optimal parameters are displayed with a roughly estimated 0.2% relative error because of the reasons discussed in Section 4.2	34
4.5	Mesh Grid run ($\sim 10^7$ MC steps) on an $N = 2$ system governed by a $\omega = \omega_1$ harmonic trap and a $\tilde{\alpha} = 0.05$ Lennard-Jones potential. $p[0]$ is the α parameter; $p[1]$ is the β_{AB} parameter. The minimum-energy point is highlighted in red both in the energy plot and in the variance plot. It is evident how the variance valley does not mimic the energy valley.	37
4.6	Comparison of Mesh Grid runs ($\sim 10^7$ MC steps) on the $N = 4$ systems governed by a $\omega = \omega_1$ harmonic trap. The top figures depict a Mesh Grid slice of the $\tilde{\alpha} = 1.00$ system; the middle figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.40$ system; the bottom figures depict a Mesh Grid slice of the $\tilde{\alpha} = 0.05$ system. $p[0]$ is the α parameter; $p[1]$ is the β_{AA} parameter; $p[2]$ is the β_{AB} parameter. The minimum-energy point is highlighted each energy plot and in the corresponding variance plot. It is evident how the variance valley does not strictly mimic the energy valley.	38
4.7	Partial energetic balances per particle of the strongly-confined interacting systems vs. the number of particles N . The two-body benchmark reference is also marked in magenta. FinalMC ran with $2^{24} \sim 10^7$ MC cycles.	39
4.8	Radial one-body densities ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. The plots are normalized by the number of particles N	40

4.9	Radial pair correlation function ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. . . .	40
4.10	Radial one-body density ($\sim 10^7$ MC steps) of the $N = 8$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.05$ Lennard-Jones potential. OBDs of the A particles and B particles are also individually plotted. All plots are normalized by the same number of particles $N = 8$. It is evident that the OBD of species A perfectly overlaps that of species B , and each contributes exactly half of the total $\rho(r)$ as expected (Subsection 2.4.1).	41
4.11	Radial pair correlation functions ($\sim 10^7$ MC steps) of the $N = 8$ system governed by the $\omega = \omega_1$ harmonic trap and the $\tilde{\alpha} = 0.05$ Lennard-Jones potential.	41
4.12	Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap. The plots are normalized by the number of particles. It is evident how a stronger Lennard-Jones interaction between different species allows for an overall more compact system. . .	42
4.13	Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of the $N = 8$ systems governed by the $\omega = \omega_1$ harmonic trap.	42
4.14	Best energy and α values obtained for the interacting systems inside a $\omega = \omega_2 = 0.027951$ harmonic confinement.	43
4.15	Comparison of Nelder-Mead optimization processes for the $N = 2$ interacting system inside a $\omega = \omega_2$ harmonic confinement. The blue optimization utilized Eq. 2.19 to evaluate the local kinetic energy; the orange one utilized Eq. 2.20; the green one utilized Eq. 2.21. Parameters $\mathbf{p}[0]$, $\mathbf{p}[1]$ and $\mathbf{p}[2]$ correspond to α , β_{AA} and β_{AB} , respectively.	43
4.16	Partial energetic balances per particle of the weakly-confined interacting systems vs. the number of particles N . The two-body benchmark reference is also marked in magenta. FinalMC ran with $2^{24} \sim 10^7$ MC cycles.	45
4.17	Radial one-body densities ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_2$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. The plots are normalized by the number of particles.	47
4.18	Radial pair correlation function ($\sim 10^7$ MC steps) of the systems governed by the $\omega = \omega_2$ harmonic trap and the $\tilde{\alpha} = 1.00$ Lennard-Jones potential. . . .	47
4.19	Summary of the energy and α values obtained for the interacting systems. . .	48
4.20	Summary of the β_{AA} and β_{AB} values obtained for the interacting systems. . .	49
4.21	Comparison of the radial one-body densities ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. The plots are normalized by the number of particles. All the quantum droplets exhibit an approximately equal internal structure.	51
4.22	Comparison of the radial pair correlation functions ($\sim 10^7$ MC steps) of systems with and without the $\omega = \omega_2$. All the quantum droplets exhibit an approximately equal internal structure.	51
4.23	3D visualization of the particles' positions over the course of 10 non-subsequent Metropolis steps (evenly extracted from a total of 2×10^7 steps) for the $N = 12$ non-confined system. The last configuration is highlighted, while the 9 preceding ones are progressively faded.	52

List of Tables

4.1	Quantum harmonic potential predictions for parameter α and ground-state energies E_0	28
4.2	Predictions for the ground-state energy of the two-body systems. E_{CM} is an analytical estimate (the same of Table 4.1); $E_{1,2}$ is obtained by discretization of domain and automatic diagonalization.	29
4.3	VMC results for non-interacting harmonic $V = \frac{1}{2}\omega^2 R^2$ systems. Different optimization methods were tested. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Both optimization algorithms used $\text{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles.	30
4.4	VMC results for interacting systems inside a $\omega = \omega_1 = 0.111803$ harmonic confinement. Different optimization methods were tested ("N-M" stands for "Nelder-Mead"; "M-G" stands for "Mesh Grid"). The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.05, 0.05, 0.05]$ as initial simplex size and 8×10^6 MC cycles. BFGS ran with 10^7 MC cycles. Adam ran 120 steps with 10^7 MC cycles and an initial learning rate of 0.03. Nelder-Mead and BFGS used $\text{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values.	35
4.5	VMC results for interacting systems inside a $\omega = \omega_2$ harmonic confinement. Different optimization methods were tested ("N-M" stands for "Nelder-Mead"; "M-G" stands for "Mesh Grid"). The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.50, 0.05, 0.05]$ as initial state setting and 8×10^6 MC cycles. BFGS ran with 8×10^6 MC cycles. Adam ran 1000 steps with 5×10^5 MC cycles, an initial learning rate of 0.05 and <code>autoMCincr.wait</code> = 500. Both NLOPT optimization algorithms used $\text{x_tol} = 10^{-5}$ as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values.	44

4.6 Successful VMC results for interacting systems without any confinement. The loss function is $\mathcal{L} = E + w \text{Var}(E)$. The initial guess parameters are indicated with the apex $\ll^{(0)}\gg$. Nelder-Mead ran with $\pm [0.50, 0.05, 0.05]$ as initial state setting, $\times 10^7$ MC cycles and used `x_tol` = 10^{-5} as a stopping criterion. FinalMC ran with $\sim 10^8$ MC cycles. Mesh Grid ran with $\sim 10^7$ MC cycles. As explained in Section 4.2, more conservative and realistic estimates of the optimal parameters would probably stop at the 3rd significant digit. The computed energies are plausibly only upper bounds on the true ground-state values. 48